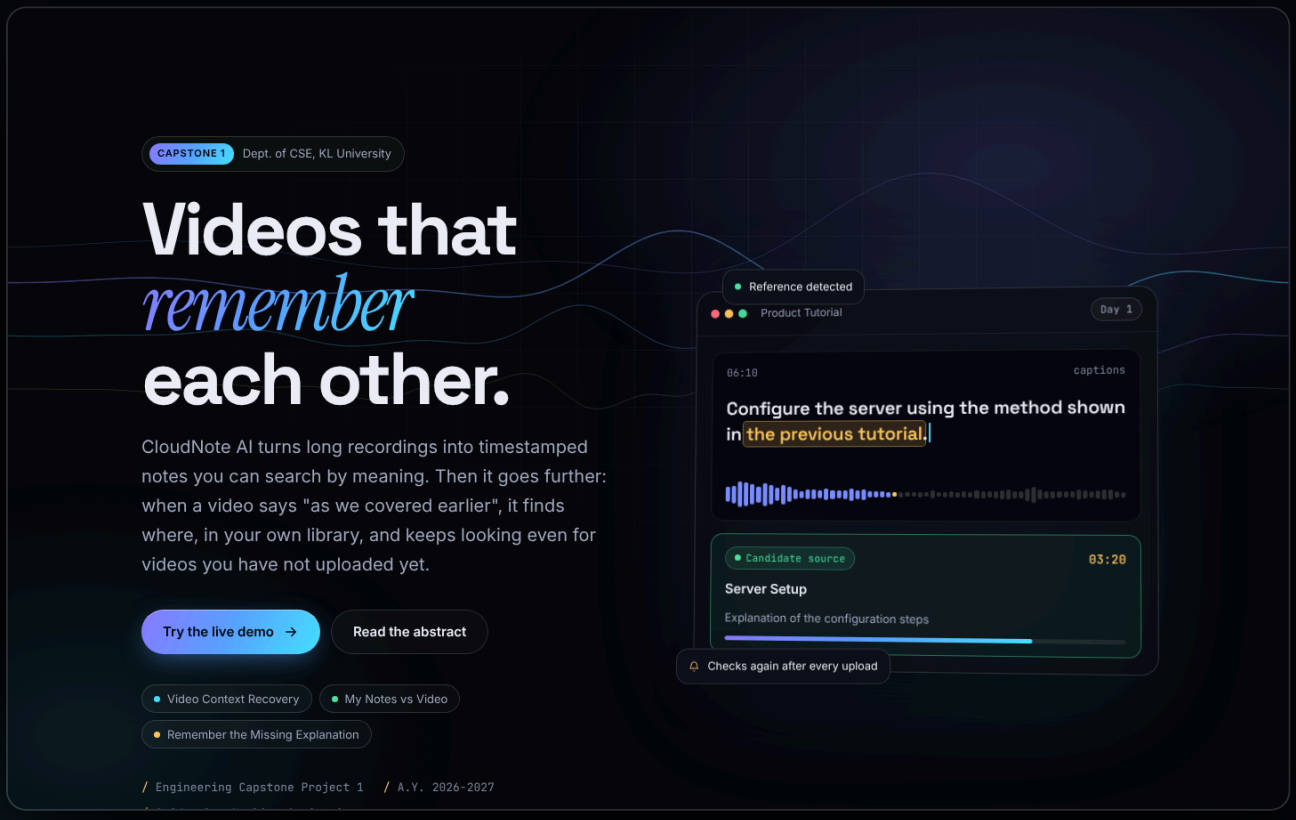
CloudNote AI

User Manual

Videos that remember each other

The complete guide to every section, control and feature of the CloudNote AI website and live demo, for visitors, presenters, reviewers and maintainers.



Capstone Project 1 · Dept. of CSE, KL University

Site build 1.0 · Manual edition 1 · October 2026

ABOUT THIS MANUAL

How to use this guide

This manual describes the CloudNote AI website and its live demo. Every screenshot was captured from the running site, using the states described in the text.

IF YOU ARE...	START WITH
A student or first-time visitor trying the demo	Chapter 3 (Getting started), Chapter 10 (The 60-second story), Chapters 6 to 9
Presenting the project	Chapter 10 (presenter script), Chapter 1 (scope) and the notes on honesty in 1.4
A reviewer or examiner	Chapters 1, 11, 12, 13 and 14 (methodology and limits)
A teammate maintaining the site	Chapter 16 (Technical reference) and Chapter 17 (Troubleshooting)

Conventions

- **Bold text** names an on-screen control exactly as it is labelled, for example **Accept this source**.
- `Monospace` marks keys, commands, file names and technical identifiers.
- Status chips: Covered · Possible mismatch · Missed key point · candidate found.
- Times such as 04:20 are positions inside a video, in minutes and seconds.

> **Important: illustrative data**
>
> No real videos are processed by the website. The three tutorials, their transcripts, timestamps, similarity scores and notes are small invented examples that mirror the worked examples in the Project Review-1 deck. They are not measured results. Where the proposed system differs from the in-browser demo, the manual says so.

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CHAPTER 1

Overview of CloudNote AI

CloudNote AI is a cloud-based platform that turns long videos into timestamped notes, searches them by meaning, and connects what one recording leaves unexplained to where another recording explains it.

3	26	4	15	13
demo videos	transcript chunks	feature tabs	papers reviewed	engine tests

1.1 The problem it addresses

Online learning, meetings, training sessions and tutorials produce hours of recordings. Watching a whole video again to find one explanation, or to check your own notes, is slow and repetitive. Three specific pains drive this project:

- **References without a source.** A video says "as discussed earlier" or "see the previous tutorial" and never repeats the explanation. The viewer has to search other recordings to find it.
- **Notes that drift from the video.** Personal notes may omit an important point or record a detail differently from the video. Checking by hand takes time.
- **Hours spent re-watching.** Making notes by hand, and hunting for one section, makes key concepts hard to spot and revisit.

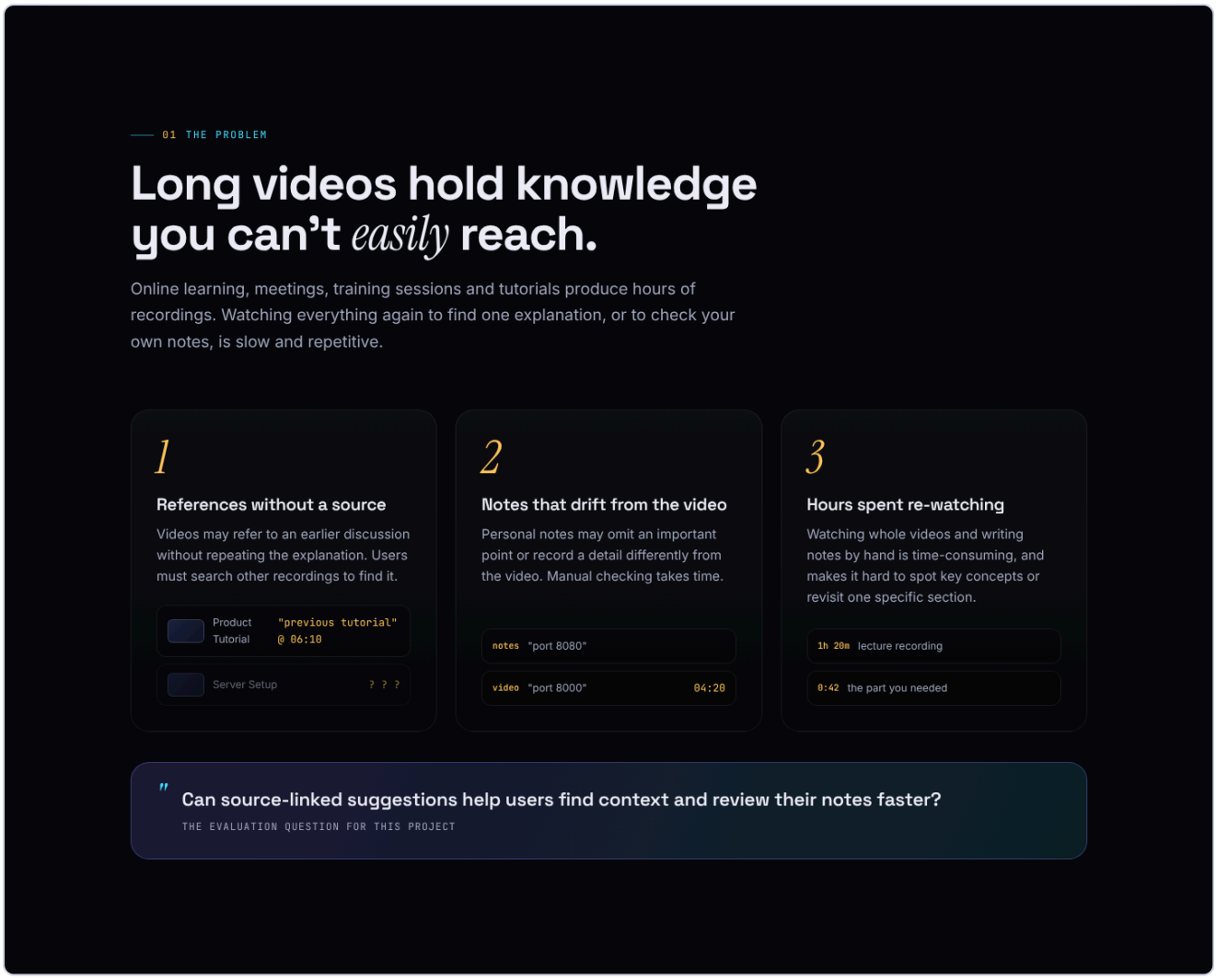


Figure 1.1 · The Problem section of the website: three pains and the evaluation question.

The project's evaluation question is: *can source-linked suggestions help users find context and review their notes faster?*

1.2 What the platform does

CAPABILITY	DESCRIPTION	WHERE ON THE SITE
Transcripts and notes	Speech-to-text produces a timestamped transcript. AI turns it into organized notes, summaries and key points, each linked to its source section.	Platform, How it works
Semantic search	Search by meaning, or ask a natural-language question, and jump to the exact moment in the video.	How it works, Architecture
Video Context Recovery	Detects phrases such as "as discussed earlier", searches your other videos and suggests up to three matching excerpts.	Live demo, Context Recovery tab
My Notes vs Video	Compares your pasted notes with the transcript: covered points, missed key points and possible mismatches, each with a timestamp.	Live demo, Notes vs Video tab

CAPABILITY	DESCRIPTION	WHERE ON THE SITE
Remember the Missing Explanation	Saves a reference that has no source yet and checks it again after every new upload, with one in-app notification.	Live demo, Missing Explanation tab
Library Map	Turns confirmed references into a dependency map, a suggested watch order and a gaps report.	Live demo, Library Map tab

1.3 Who it serves

- **Students** revisit a lecture section instead of the whole recording.
- **Professionals** get meetings and interviews with action items and decisions.
- **Educators** turn long sessions into structured, searchable material, and use the Library Map to see which explanations are still missing.
- **Organizations** keep training recordings in centralized, accessible knowledge.

1.4 Scope of the website

The website presents the project and includes a working miniature of the workspace. It is not the full application. The proposed system runs React and FastAPI with FFmpeg and Whisper on a laptop, stores data in Supabase and uses Gemini for notes. The website demonstrates the ideas with a small illustrative library of three tutorial videos.

> **Important: illustrative data**
>
> The project is at Project Review-1. Every transcript, timestamp, similarity score and example note on the site and in this manual is a small invented example that mirrors the worked examples in the project deck. They are not measured results. The site makes no accuracy, speed or usage claims.

CHAPTER 2

Key concepts

A short vocabulary used throughout the site and this manual.

TERM	MEANING
Library	The set of videos a user has uploaded. Searches only ever look inside the signed-in user's own library.
Chunk	A short stretch of transcript with a start time. Every note, search result and link points back to a chunk, so every claim has a source moment.
Timestamp	The start time of a chunk, shown as mm:ss, for example 04:20. Clicking one seeks the player to that moment.
Reference phrase	Wording in a video that points elsewhere, such as "the previous tutorial", "as we covered earlier" or "last time".
Surrounding sentences	The reference chunk plus the next chunk. They are used as the search query because they describe the missing topic.
Similarity score	A number from 0 to 1 showing how closely an excerpt matches a query. It marks a candidate, never proof of the original source.
Threshold	The minimum score for a suggestion. In the demo it is 0.30. Below it, CloudNote AI says "no suitable source yet" rather than guess.
Candidate	A suggested source excerpt waiting for the user to accept or reject it.
Pending reference	A reference with no suitable source yet, saved so it can be checked again after future uploads.
Worker	The background job that, after a new video is indexed, compares it with the user's pending references.
Key point	A short statement extracted from a video, linked to its source chunk. My Notes vs Video compares your notes against key points.
Gap	A reference that still has no confirmed source. The Library Map lists gaps as recordings that remain to be made.

2.1 The control principle

Nothing changes silently. Links are made only when the user accepts a candidate. Note edits are saved only when the user presses Save. A claim that is absent from the video is not automatically treated as false. This is a design rule of the project, and the demo follows it.

CHAPTER 3

Getting started

How to open the site, move around and use the keyboard shortcuts.

3.1 Access and devices

Open the deployed address of the site in any modern browser. No account, installation or sign-in is needed, and the live demo runs entirely in the browser. The layout adapts from phones (about 360 px wide) to large desktop screens.

DEVICE	LAYOUT
Desktop (1060 px and wider)	Full top navigation, two-column demo workspace, player stays in view while you scroll the panel
Tablet	Single-column demo, tabs scroll horizontally, menu button replaces the top links below 1060 px
Phone	Stacked sections, drawer menu, one video card per row in the demo library

3.2 Top bar



Figure 3.1 · The top bar: brand, section links, search button.

- The brand mark returns to the top of the page.
- Section links highlight the section you are reading.
- The **Search** button opens site search. A thin gradient bar at the very top shows how far down the page you are.
- A round arrow button appears after you scroll and returns you to the top.

3.3 Site search

Press `Ctrl` + `K` (`Cmd` + `K` on Mac), or `/` outside a text box, to open search. It covers sections, features, pipeline steps, technology components, all 15 papers and the FAQ, 52 entries in total. Results are ranked by the same TF-IDF method the demo uses, plus plain text matching, so partial words work.

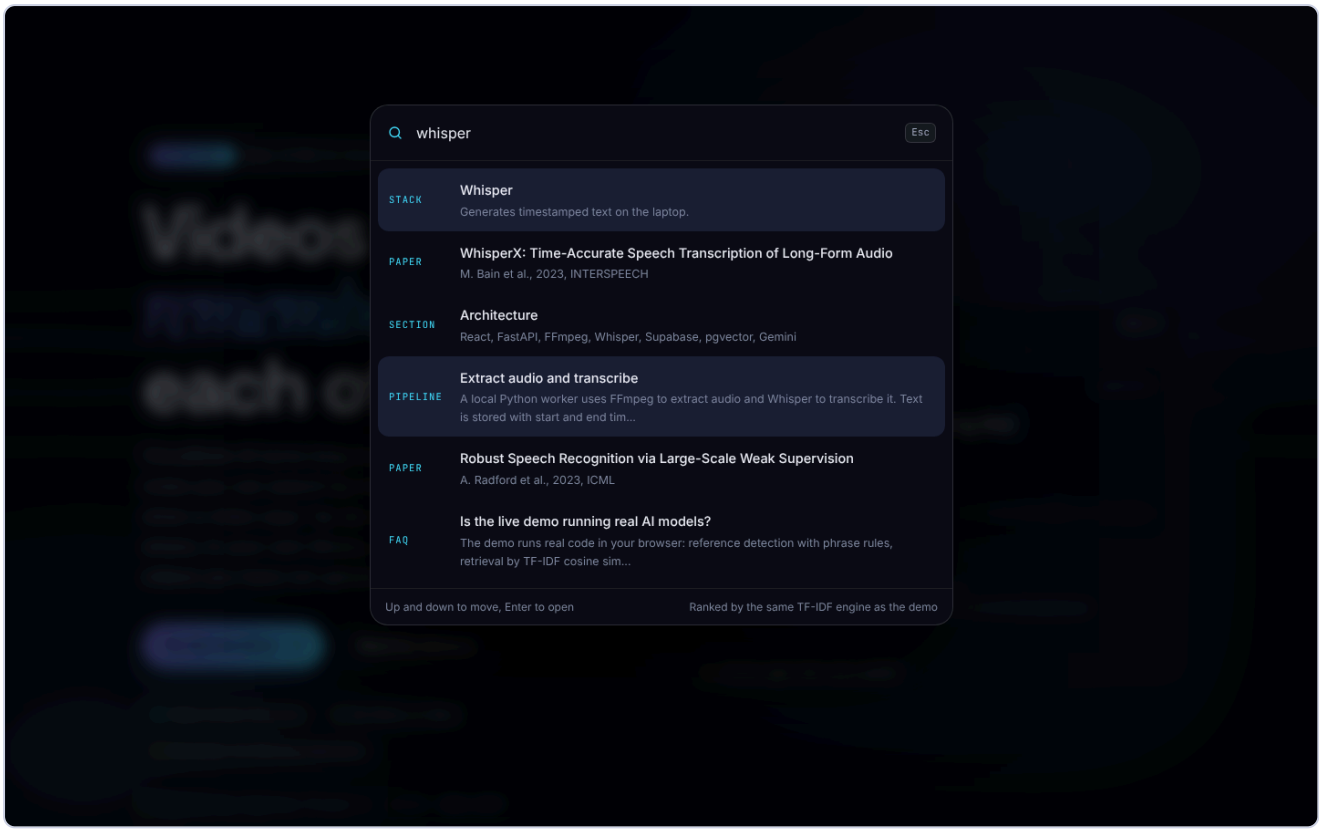


Figure 3.2 · Site search open with the query "whisper".

KEY	ACTION
Ctrl/Cmd + K or /	Open site search
Up / Down	Move through results
Enter	Open the selected result and scroll to it
Esc	Close search, or close the mobile menu or notification list

3.4 Phones

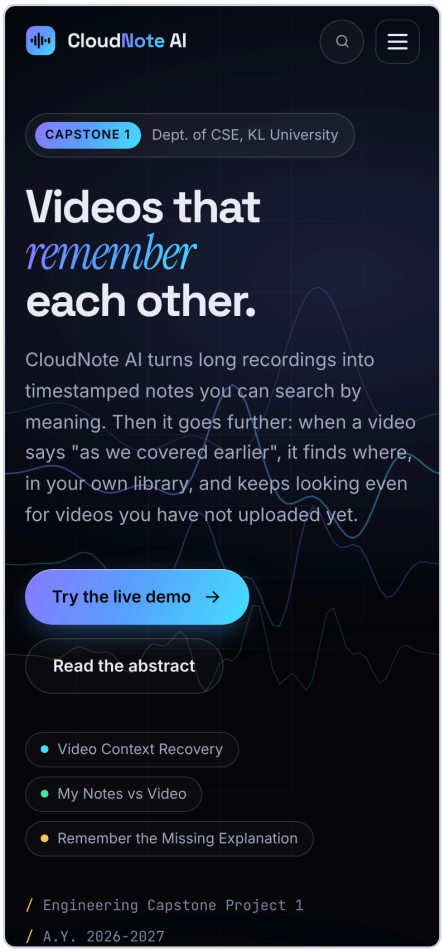


Figure 3.3 · Phone: hero.

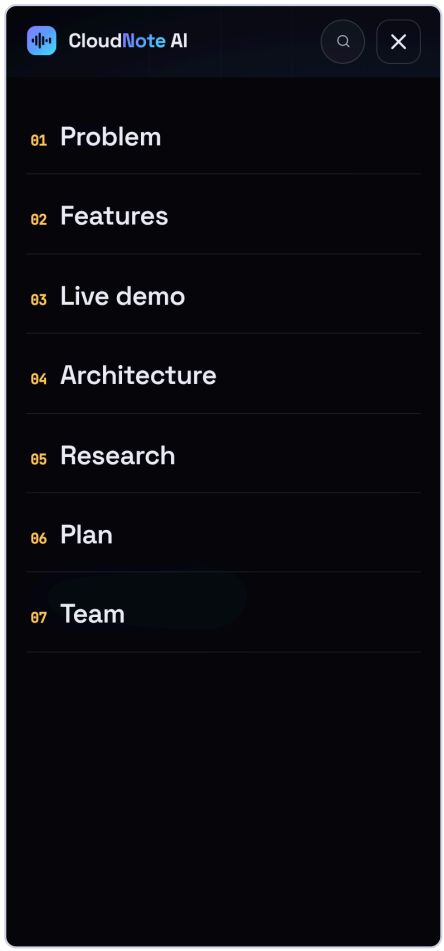


Figure 3.4 · Phone: menu drawer.

3.5 Accessibility

- A "Skip to content" link appears on the first Tab press.
- Every control is reachable by keyboard with a visible focus ring. Tabs, the pipeline steps and the seek bar respond to arrow keys.
- If your device is set to reduce motion, animations are replaced by still states and the page scrolls without easing.
- Text and controls use high-contrast colors on a dark background. Status is never shown by color alone: every chip also has a text label.

CHAPTER 4

The home page, section by section

The page is one scroll. This chapter walks through each section in order.

4.1 Hero

The first screen states the idea: *videos that remember each other*. On the right, a short looping animation shows three moments: a reference detected and matched (Product Tutorial, 06:10), a reference with no source that is remembered, and the next-day upload that answers it. Two buttons lead to the live demo and the abstract. A scrolling strip below lists the technologies in the proposed stack.

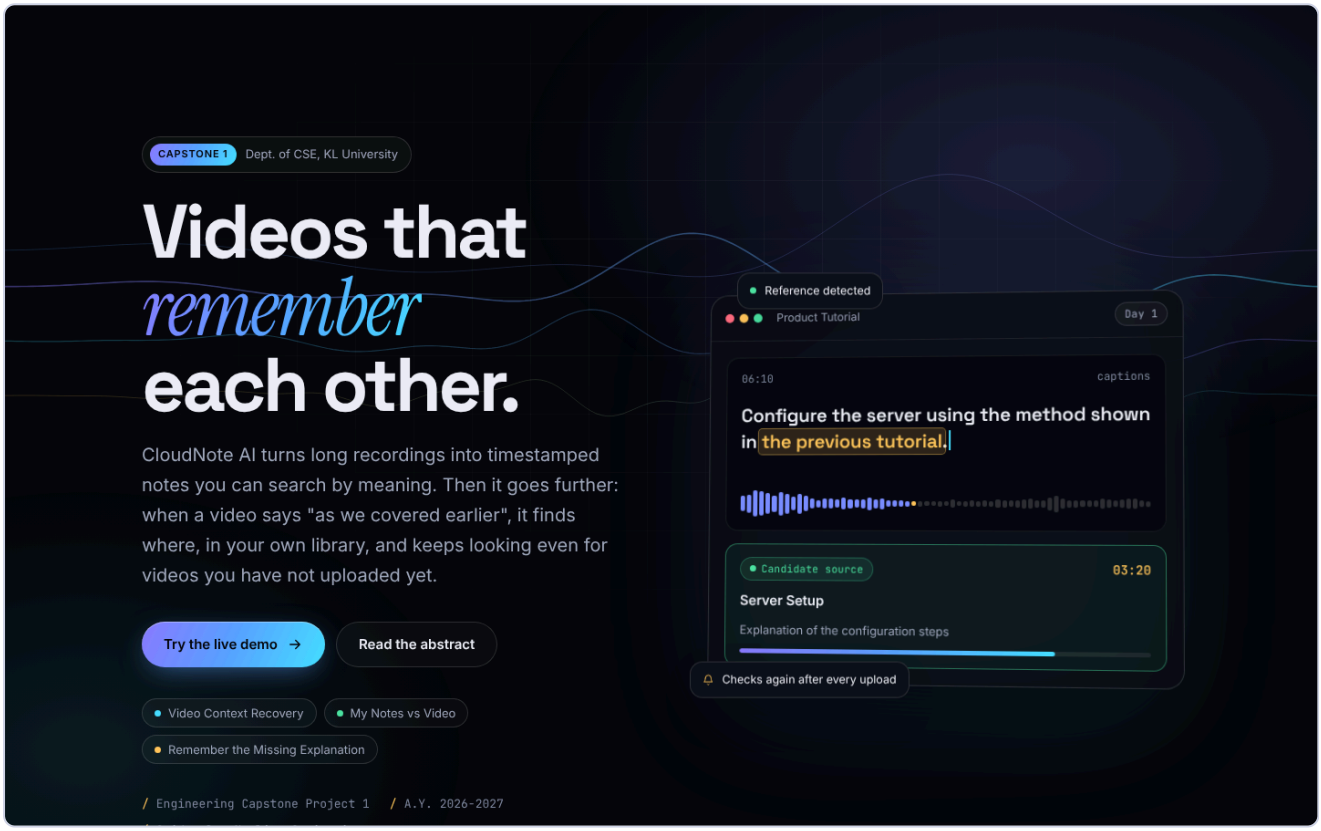


Figure 4.1 · The hero with the animated example.

4.2 Problem and Platform

The Problem section (Figure 1.1) lists three pains. The Platform section summarizes the core capabilities from the project abstract: speech to transcript, notes and summaries, important timestamps, action items and decisions, semantic search and cloud storage.

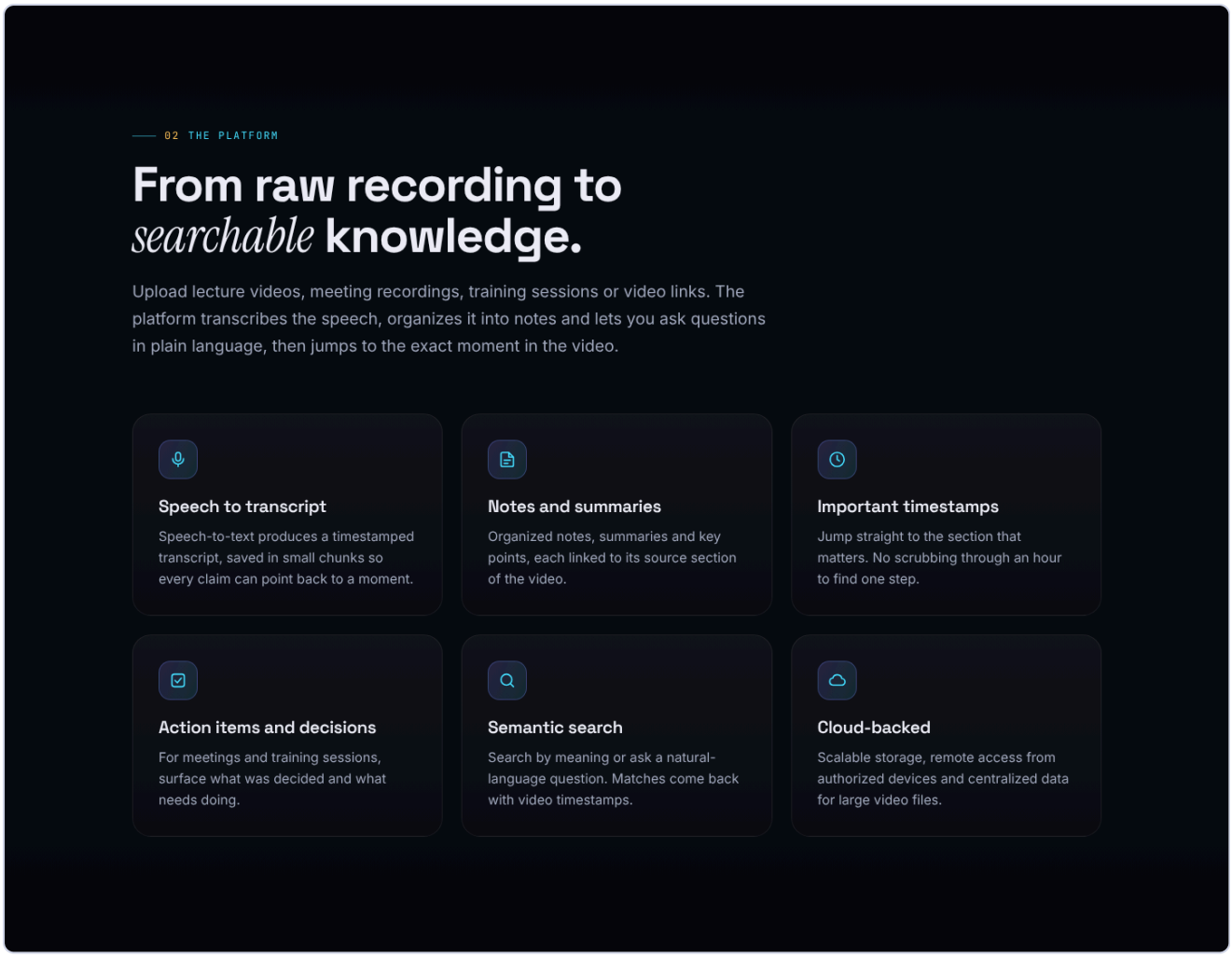


Figure 4.2 · The Platform section: six core capabilities.

4.3 Features

Three cards introduce the three extensions, each with a **Try it live** button that opens the matching demo tab. A wider card introduces the Library Map.

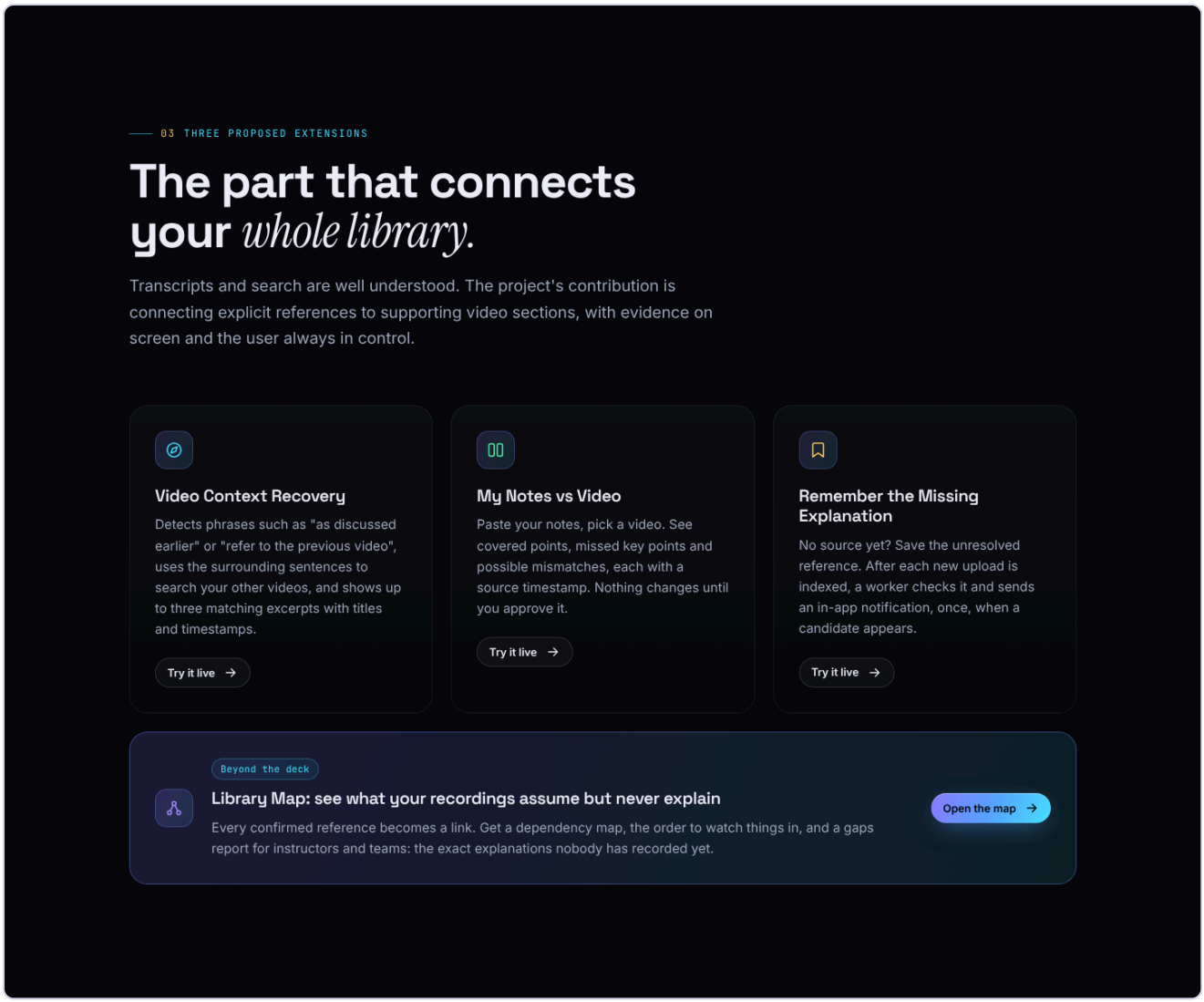


Figure 4.3 · The Features section.

4.4 Live demo, How it works, Architecture

These are covered in detail in Chapters 5 to 11. Chapter 5 begins the live demo.

4.5 Research, Plan, Roadmap, FAQ

Research lists the 15 papers and the research gap (Chapter 12). Plan gives the day estimates, evaluation checks and free-tier limits (Chapter 13). Beyond the demo lists the intended audience and future enhancements. The FAQ answers the questions a reviewer is most likely to ask.

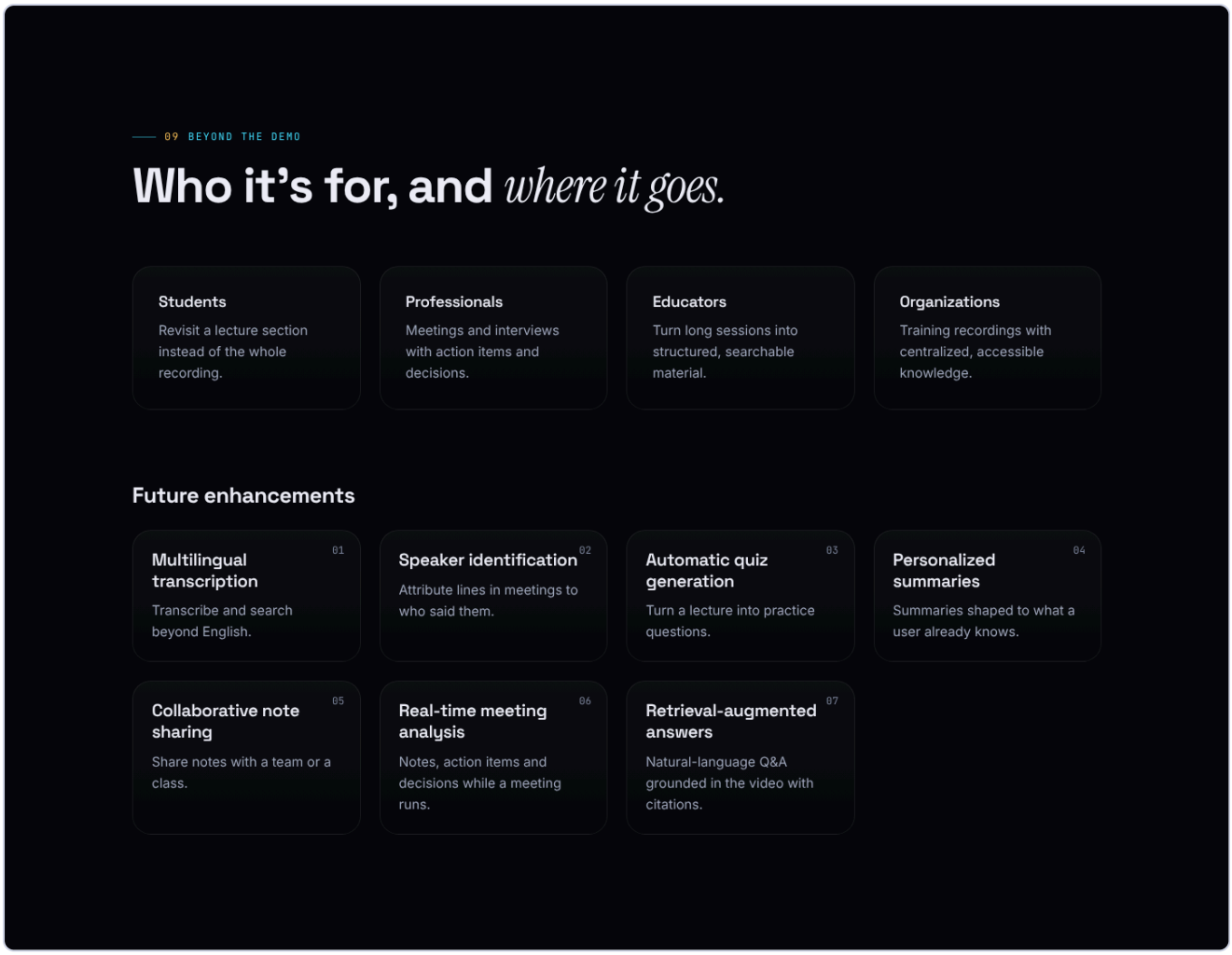


Figure 4.4 · Audience and future enhancements.

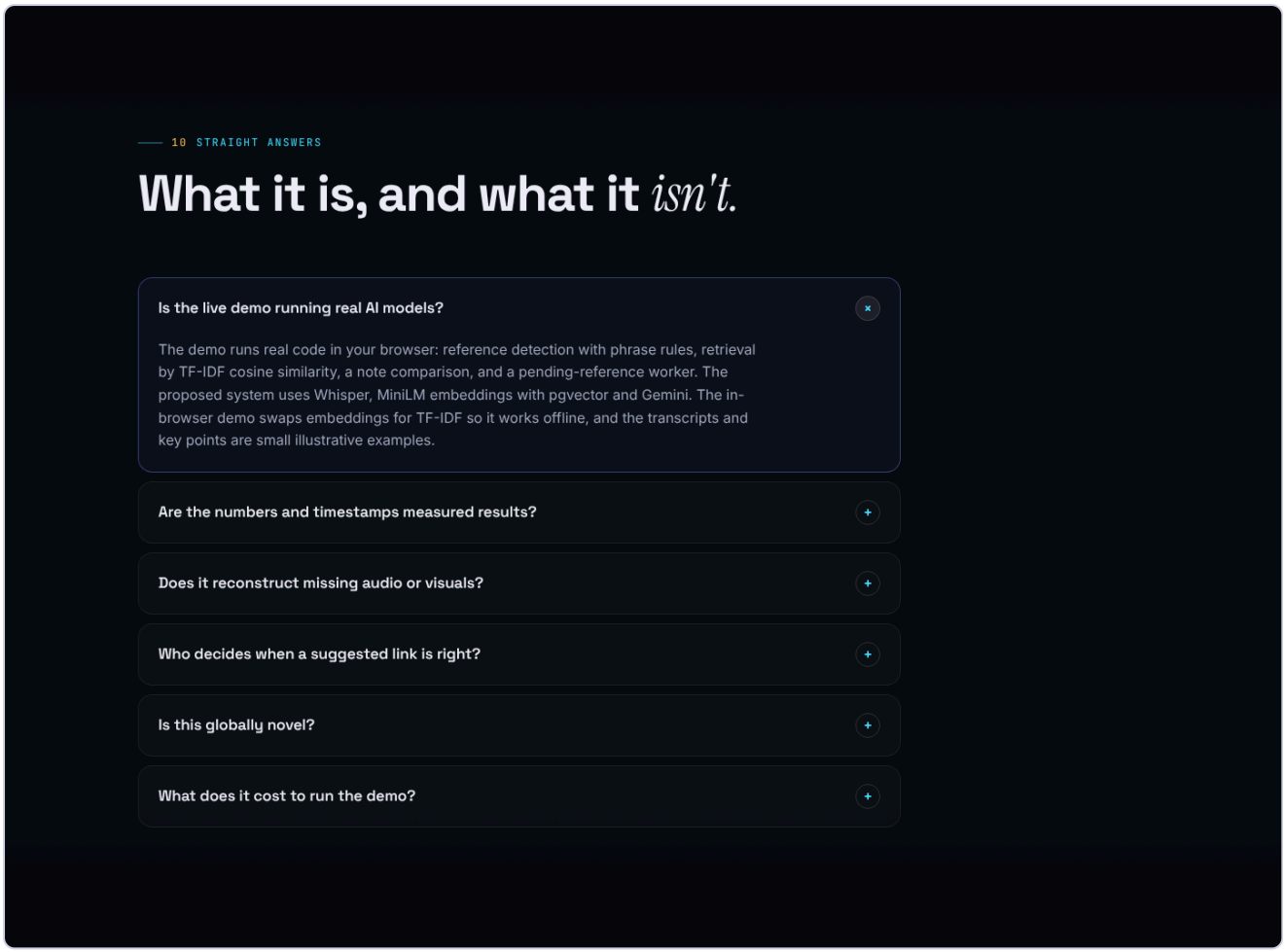


Figure 4.5 · The FAQ: straight answers about what is real and what is illustrative.

4.6 Team and References

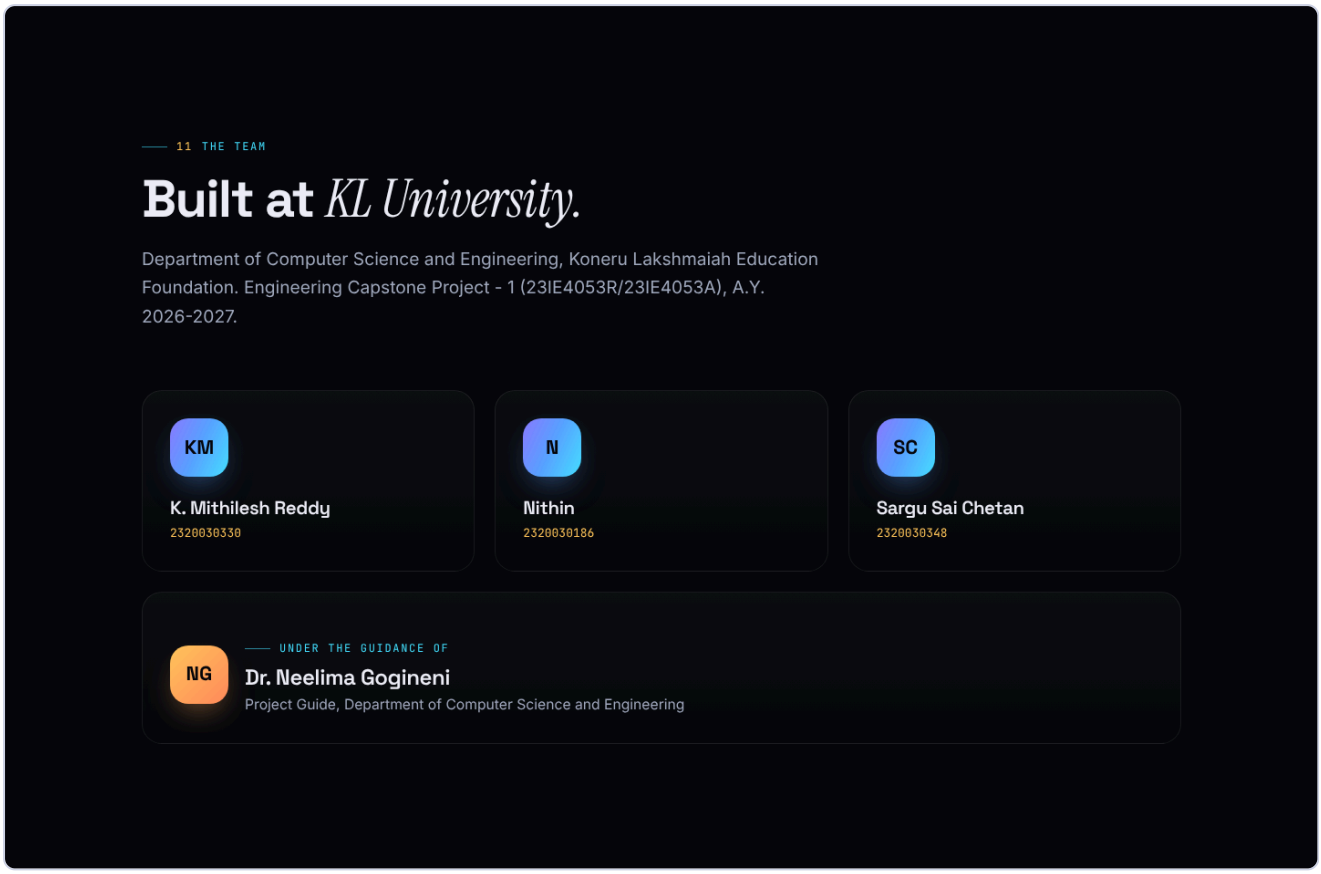


Figure 4.6 · The team and project guide.

The team section lists the three students and the project guide. References lists all 15 papers with DOI links and the eight implementation references.

CHAPTER 5

The live demo workspace

A working miniature of CloudNote AI that runs in your browser. This chapter explains the shared parts: the library, the player and the notification bell.

5.1 Layout

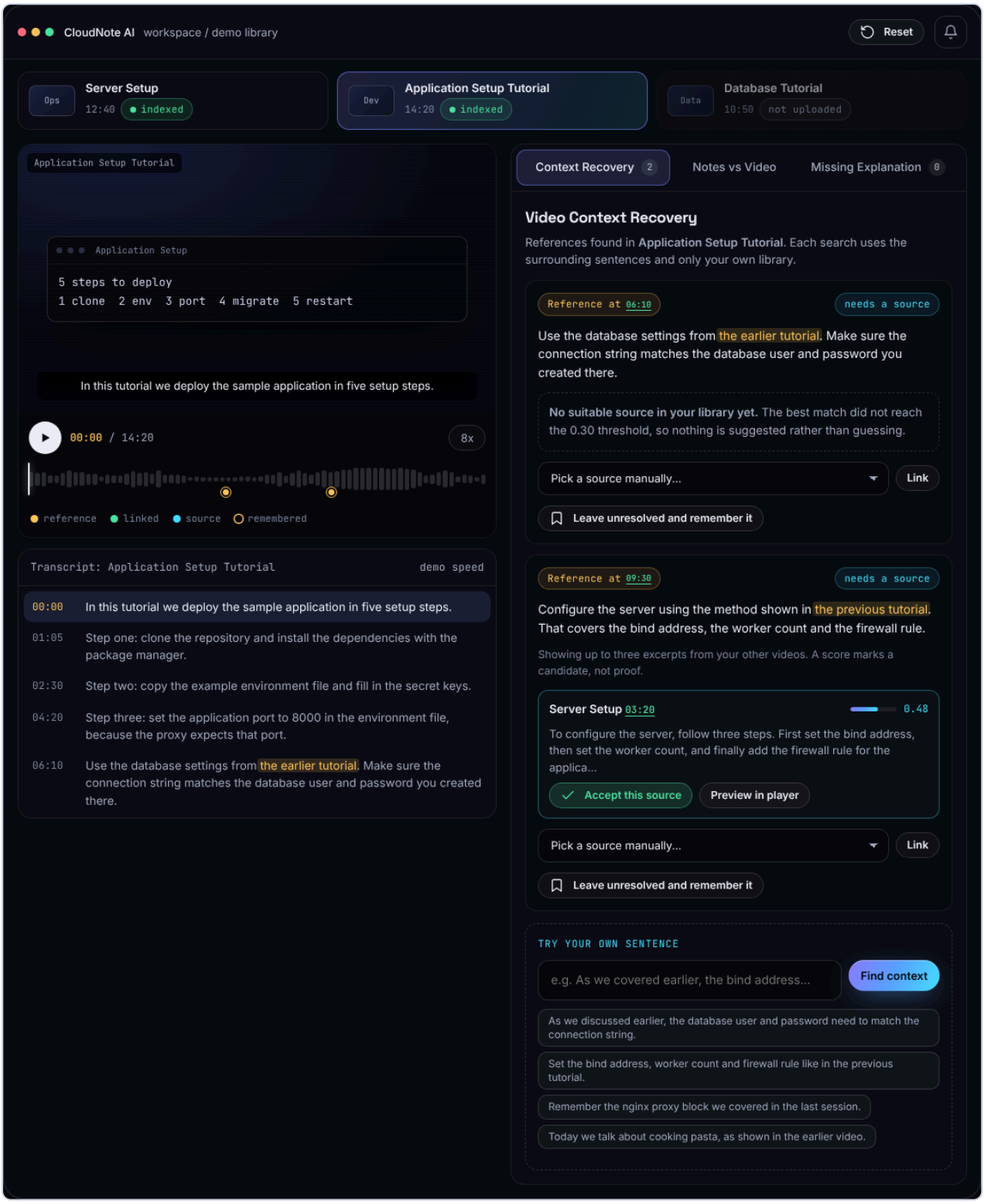


Figure 5.1 · The workspace on first load.

1. **Title bar.** Reset button and the notification bell.
2. **Library strip.** Three videos. Click one to load it in the player.
3. **Player** with simulated video, captions, controls and a waveform seek bar.
4. **Transcript** for the selected video.

5. **Feature panel** with four tabs on the right.

5.2 The library

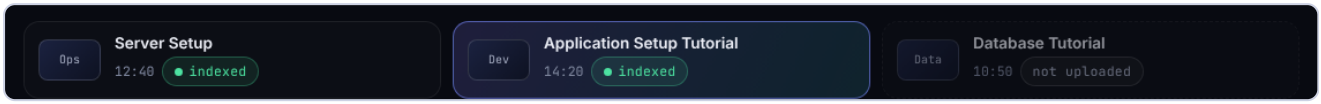


Figure 5.2 · The library strip: two indexed videos and one not yet uploaded.

VIDEO	LENGTH	TAG	STATE AT START	ROLE IN THE DEMO
Server Setup	12:40	Ops	Indexed	The earlier tutorial. Its 03:20 section explains the server configuration.
Application Setup Tutorial	14:20	Dev	Indexed	The video being worked on. It contains two references and the notes to compare.
Database Tutorial	10:50	Data	Not uploaded	Arrives "tomorrow". Its 04:20 section explains the database settings.

Clicking the not-yet-uploaded video explains that it arrives tomorrow and takes you to the Missing Explanation tab, where you can upload it.

5.3 The player



Figure 5.3 · The player: simulated screen, caption, controls and seek bar with pins.

CONTROL	WHAT IT DOES
Play / Pause	Plays the video at demo speed. Playback stops at the end.
Speed chip	Cycles 4x, 8x (default) and 16x. The demo is deliberately faster than real time.
Waveform	Click or drag to seek. With the bar focused: Left/Right move 10 s (Shift: 30 s), Home and End jump to the ends, Space plays or pauses.
Transcript line	Click any line to seek there. The current line is highlighted and scrolls into view.
Amber pin	A reference phrase was detected at this time.
Green pin	That reference has been linked.
Cyan pin	A linked source in the video you are viewing.
Hollow ring	A reference that is remembered and waiting.

The picture inside the player is a stylized terminal that changes with each transcript chunk, so you can see that the video moves on. Captions show the current chunk, with reference phrases underlined.

5.4 Notifications and reset

The bell in the title bar shows a badge for each unread notification. Notifications appear when a later upload answers a remembered reference (Chapter 8). **Reset** returns the whole demo to its starting state and clears the notes saved in your browser.



Figure 5.4 · The title bar with the notification list open (detail).

CHAPTER 6

Video Context Recovery

Find where a video's "as discussed earlier" is explained, using only your own library.

6.1 What you see

The first tab lists the references found in the Application Setup Tutorial. Each card shows the sentence with the reference phrase highlighted, its timestamp and a state chip: **needs a source**, **linked** or **remembered**.

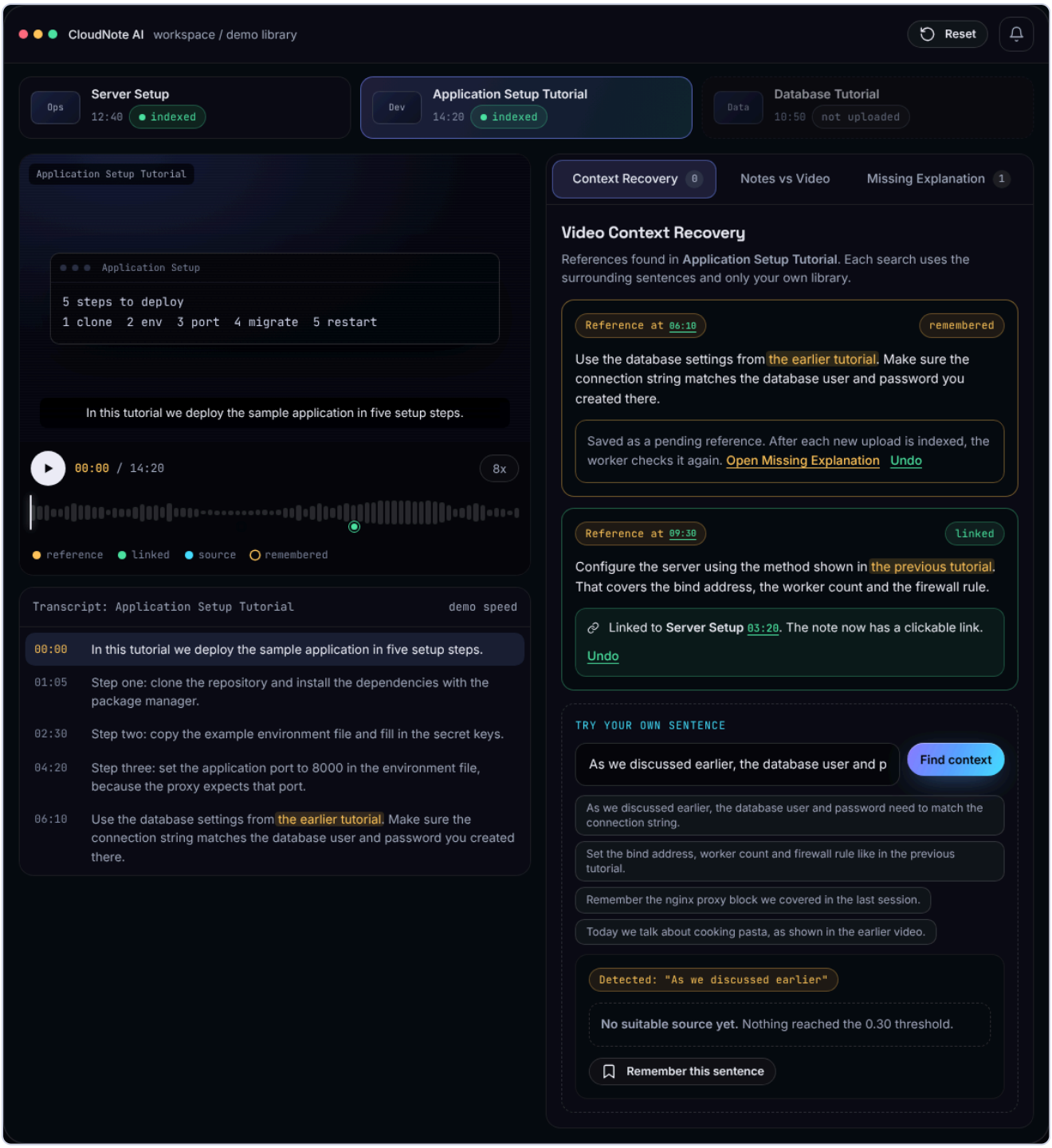


Figure 6.1 · Context Recovery after linking one reference and remembering the other.

6.2 How a reference is handled

1. **Detect.** Phrase rules find wording such as "the previous tutorial" or "as we covered earlier" (Chapter 14 lists them).
2. **Search.** The reference chunk plus the next chunk become the query. The search covers your other videos only.
3. **Suggest.** Excerpts at or above the 0.30 threshold are shown, up to three, with video title, timestamp, score bar and excerpt. If none reach the threshold the card says no suitable source yet.
4. **Decide.** You choose one of four actions.

ACTION	RESULT
Accept this source	The reference becomes linked. A green pin appears, and a cyan pin marks the source in the other video.
Preview in player	Opens the candidate in the player so you can check it before accepting.
Pick a source manually	Choose any excerpt from your other videos from the list and press Link.
Leave unresolved and remember it	Saves the reference as pending, to be checked after future uploads.

Use **Undo** on a linked or remembered card to put it back to "needs a source".

6.3 Try your own sentence

At the bottom of the tab, type any sentence containing a reference phrase and press **Find context**, or tap one of the sample chips. The result lists the detected phrase and the best matches. Pressing **Remember this sentence** saves it as your own pending reference so you can test it against tomorrow's upload.

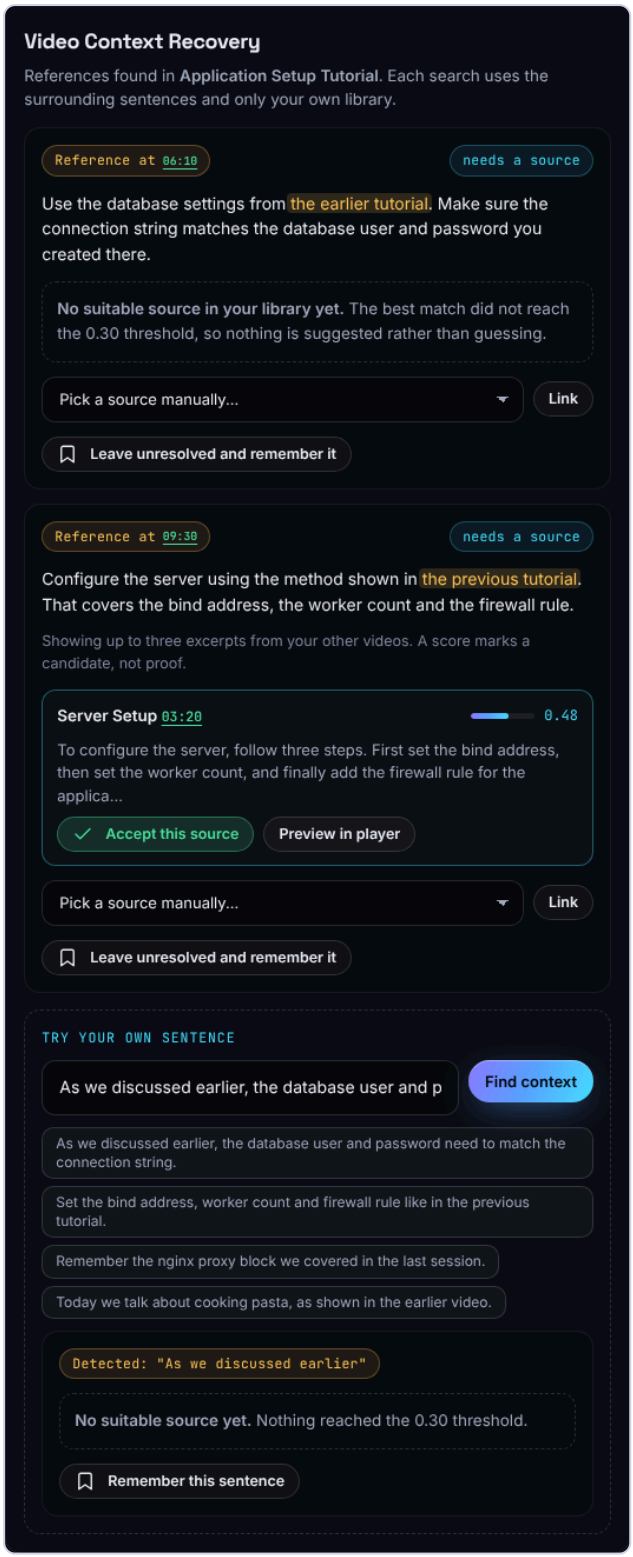


Figure 6.2 · Trying a sentence. On day 1 it finds no source above the threshold, so it offers to remember it.

Why no guess?

A tool that always returns its best match would sometimes point you at the wrong video and look confident doing it. CloudNote AI prefers an honest "not yet" and lets the worker try again when more of your library exists.

CHAPTER 7

My Notes vs Video

Check your own notes against a video's transcript, and fix gaps with evidence in front of you.

7.1 Input

The text box starts with four sample notes written from the Application Setup Tutorial. Edit them freely or paste your own, one point per line. Bullets and numbers at the start of a line are ignored.

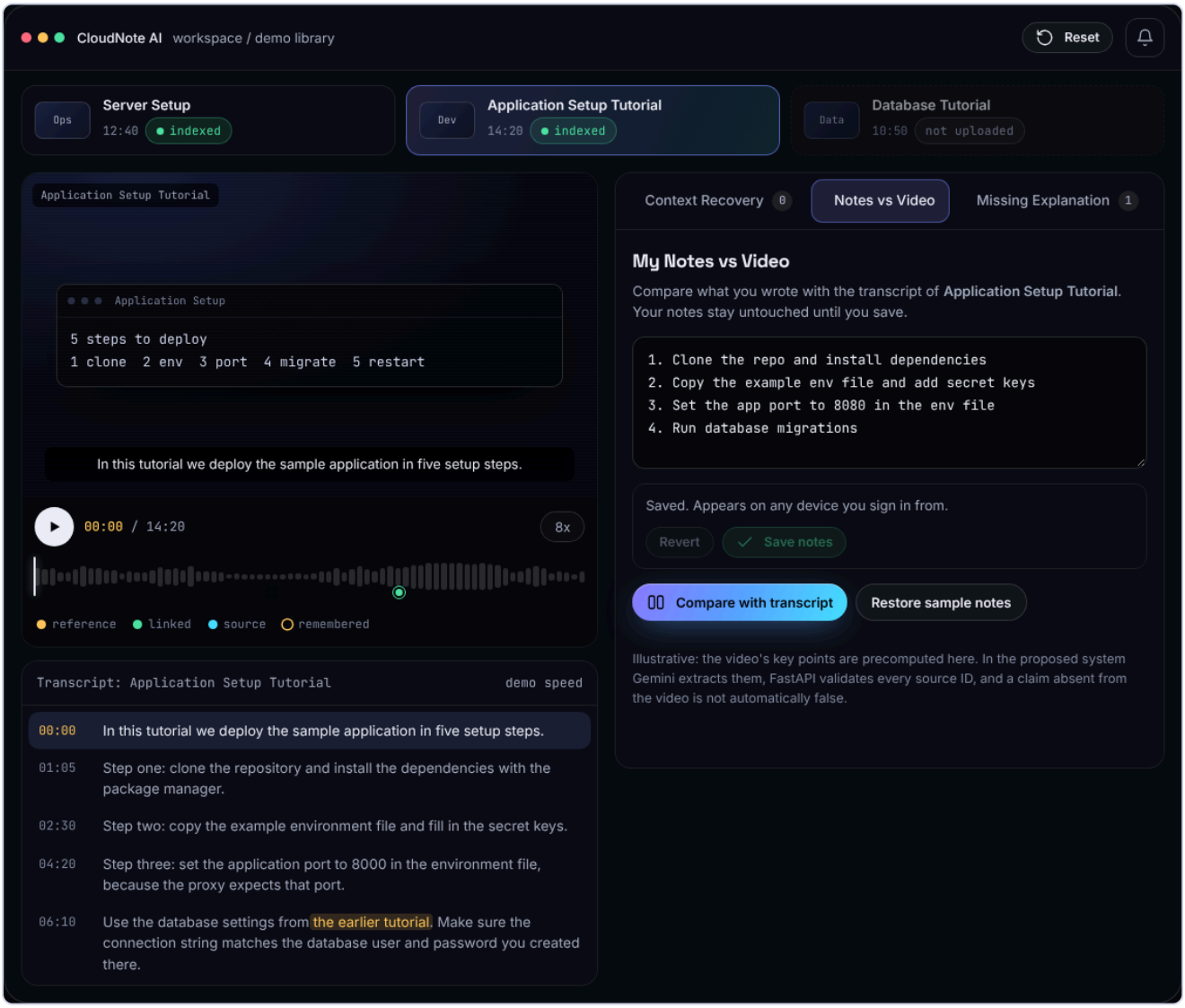


Figure 7.1 · The Notes vs Video tab before comparing.

7.2 Comparing

Press **Compare with transcript**. The video has five key points. Each is matched to the closest line of your notes. A line is treated as matching at a similarity of 0.34 or more.

VERDICT	MEANING	SAMPLE RESULT
Covered	Your notes contain the point, with no conflicting number.	Clone and install; copy the env file; run migrations
Possible mismatch	Your notes match the point but a number differs from the video.	Notes say port 8080, video says 8000 at 04:20
Missed key point	No line of your notes matches the point.	Restart the service, at 08:15

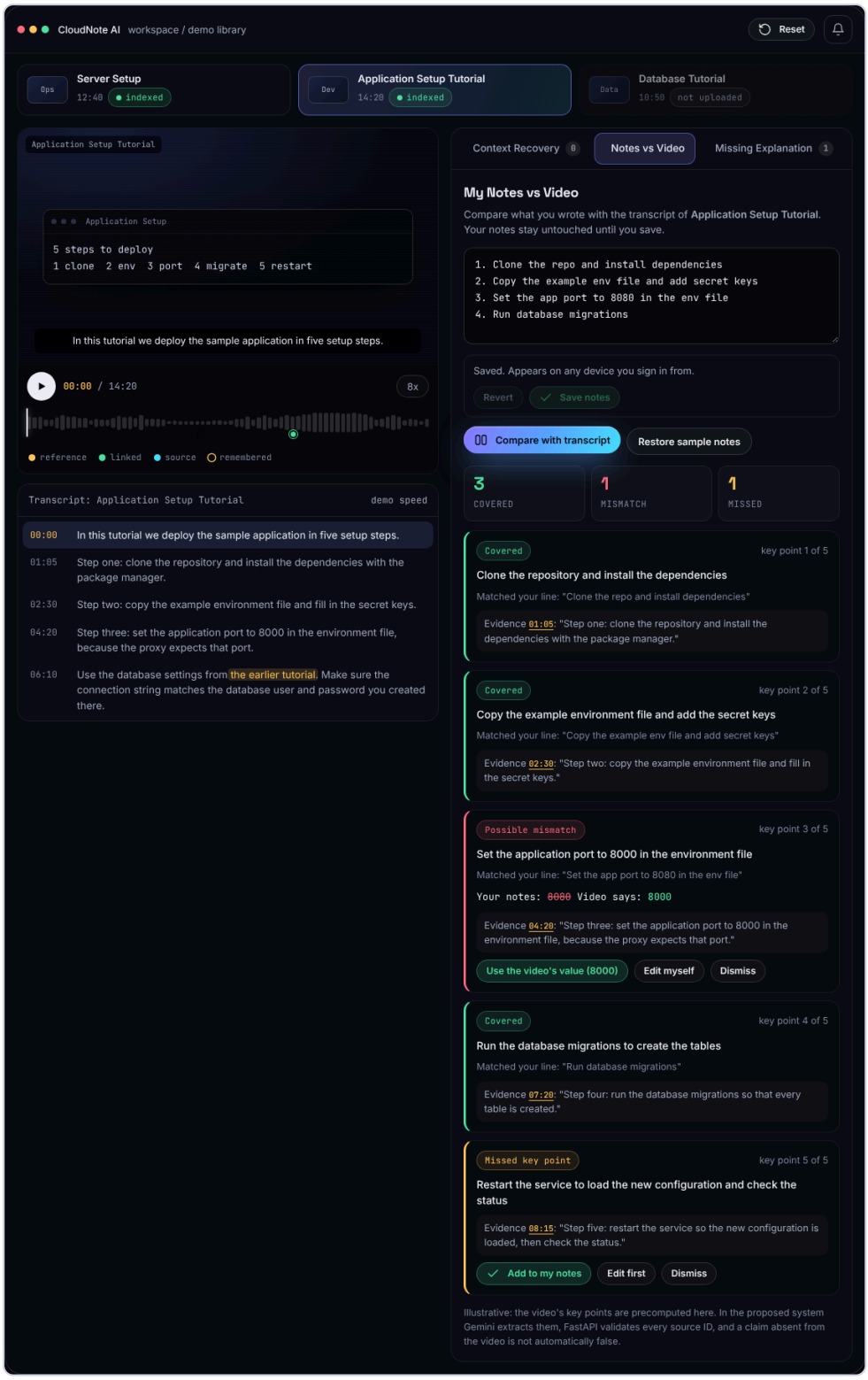


Figure 7.2 · Comparison results: 3 covered, 1 possible mismatch, 1 missed.

Each card shows the key point, the line of your notes it matched, and an evidence line with a clickable timestamp that opens the supporting moment in the player.

7.3 Acting on a result

BUTTON	EFFECT
Use the video's value	Replaces the differing number in your matching line (for example 8080 becomes 8000).
Add to my notes	Appends the missed key point as a new line, numbered if your notes are numbered.
Edit myself / Edit first	Moves the cursor to the notes box so you can word it your own way.
Dismiss / Undo dismiss	Sets a suggestion aside without changing your notes.

After each change the comparison refreshes. Fixing the port and adding the restart step turns the summary into 5 covered, 0 mismatch, 0 missed.

7.4 Saving

Changes stay unsaved until you press **Save notes**. The bar under the box turns amber while there are unsaved changes, and **Revert** restores the last saved text. Saved notes are kept in your own browser. **Restore sample notes** reloads the original four lines.

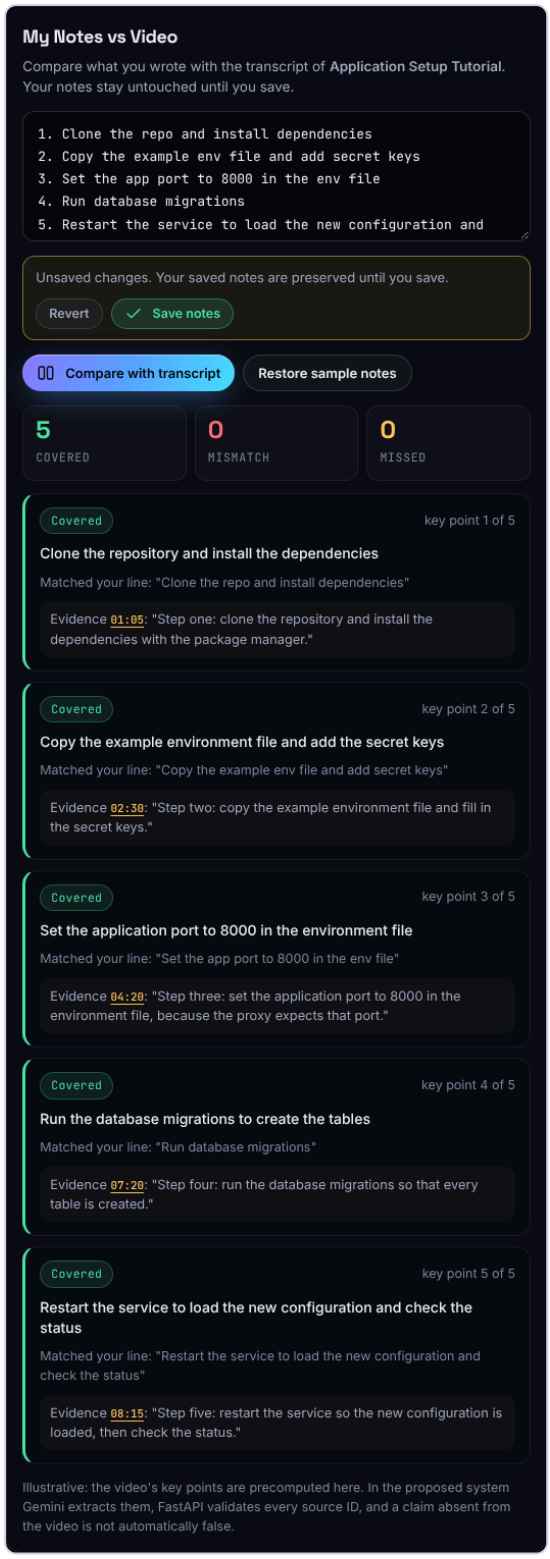


Figure 7.3 · After applying both suggestions: ready to save.

> **In the proposed system**
>
> Gemini extracts the key points with source chunk IDs, FastAPI accepts only IDs that exist in the transcript and reads timestamps from stored chunks, and Supabase stores your notes so they appear when you sign in on another device. In this demo the key points are precomputed.

CHAPTER 8

Remember the Missing Explanation

Save a question no video answers yet, and be told once when a later upload does.

8.1 The idea in five steps

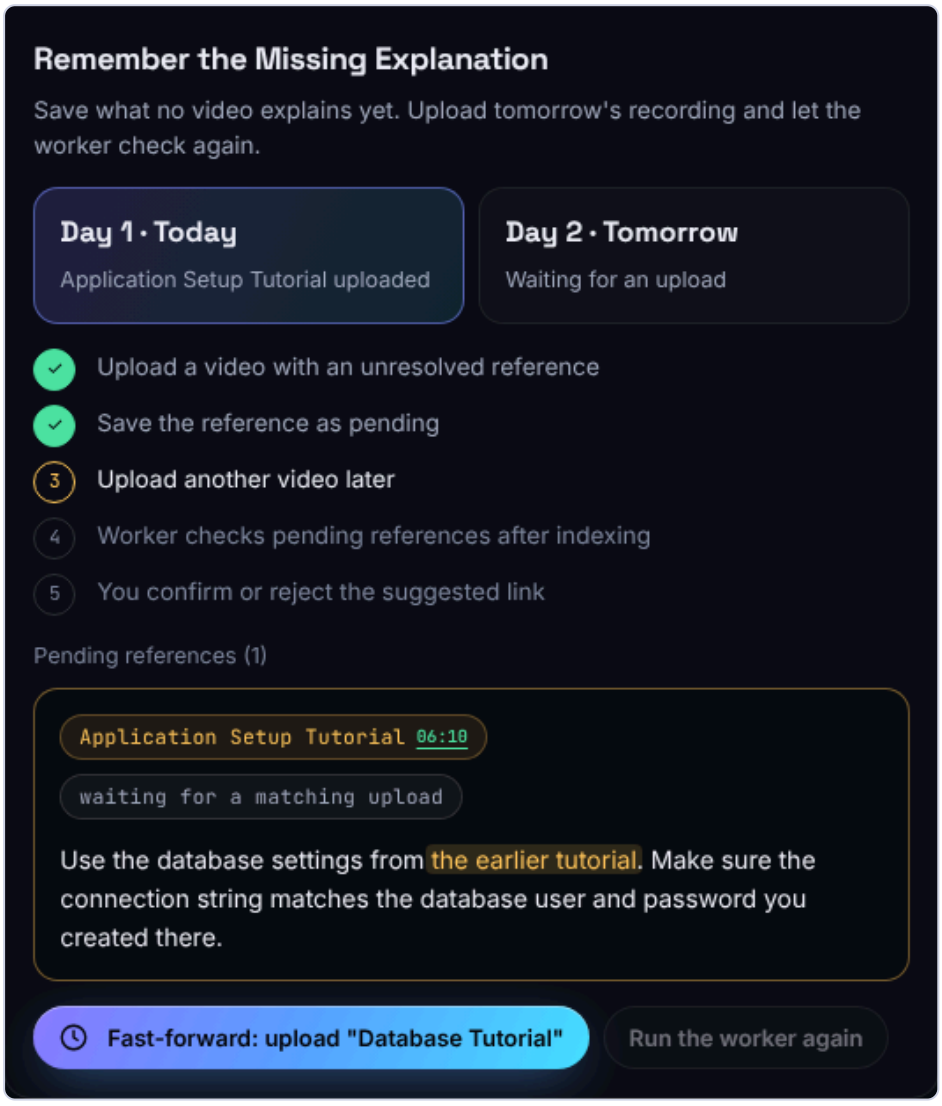


Figure 8.1 · The Missing Explanation tab at the start (nothing remembered yet).

1. A video has a reference with no suitable source.
2. You save it as a pending reference (from Context Recovery, or with the button in this tab).
3. Later you upload another video.
4. After it is indexed, the worker checks only the new video against your pending references.
5. If a match reaches the threshold you get one notification, and you confirm or reject the link.

8.2 Fast-forward to tomorrow

Press **Fast-forward: upload "Database Tutorial"**. A log shows the stages: upload, audio extraction, transcription, embedding of 6 chunks, and the worker checking pending references against that video only. The Day 2 box lights up, the video becomes available in the library, and if a candidate is found the bell rings.

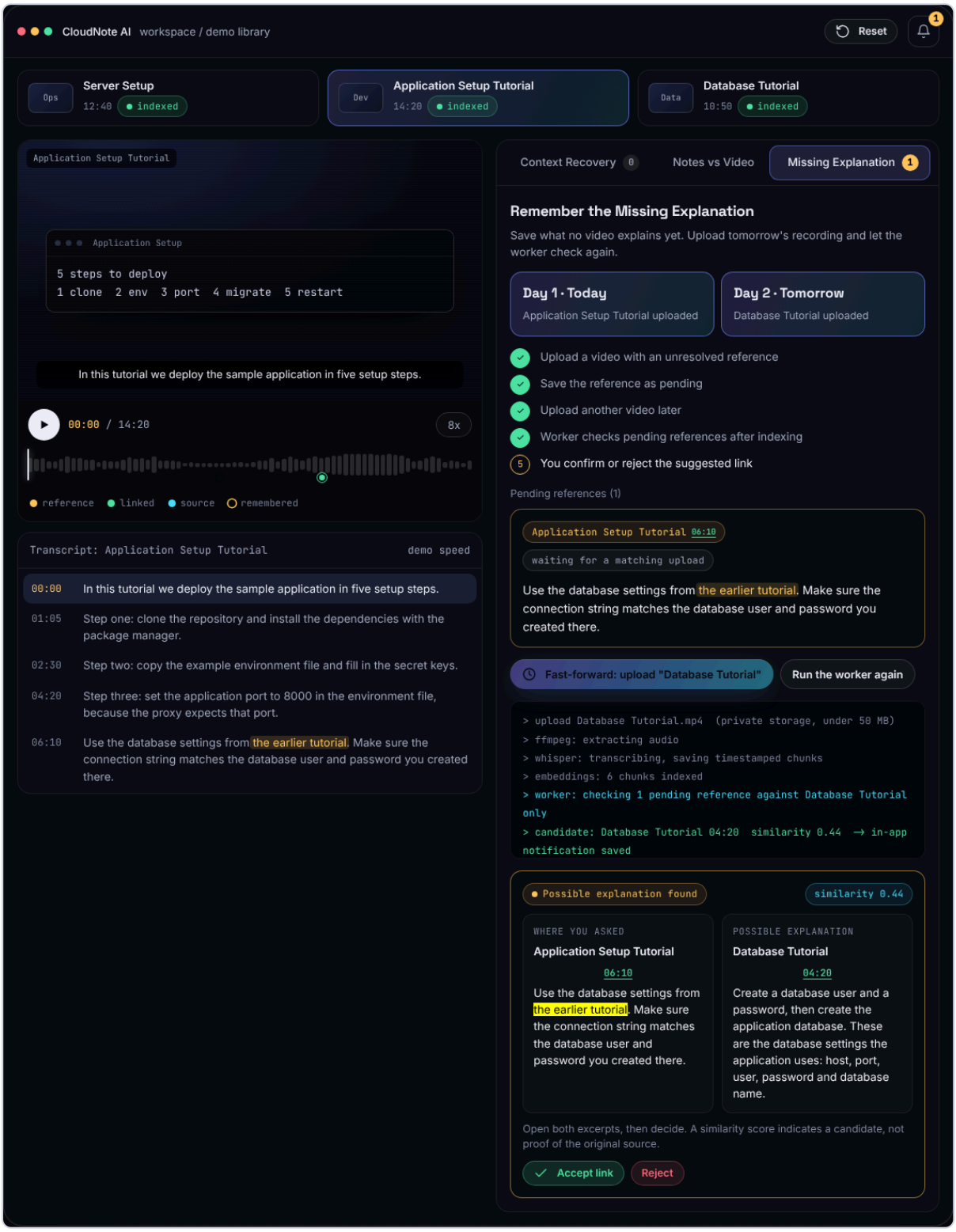


Figure 8.2 · After the upload: a possible explanation found, with both excerpts side by side.

8.3 Reviewing a candidate

The review card shows where you asked (Application Setup Tutorial, 06:10) next to the possible explanation (Database Tutorial, 04:20) with the similarity score. Click either timestamp to play that moment. Then choose:

BUTTON	EFFECT
Accept link	The reference becomes linked. The link is saved in the older video's notes, the pin turns green, and the Library Map updates.
Reject	That candidate is never suggested again. The reference keeps waiting for other uploads.

A score is a candidate, not proof. Always open both excerpts before accepting.

8.4 No repeat alerts

Press **Run the worker again** before you decide, and the log reports that no new candidates were found, because that candidate was already alerted. After you have accepted the link, it reports that nothing is pending. In the proposed system the no-repeat rule is enforced by a unique constraint on the reference and candidate pair.

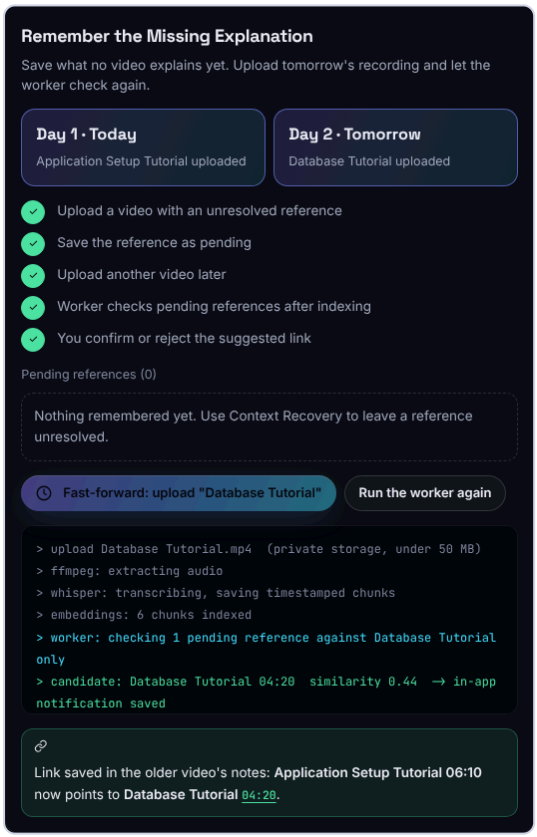


Figure 8.3 · After accepting the link.

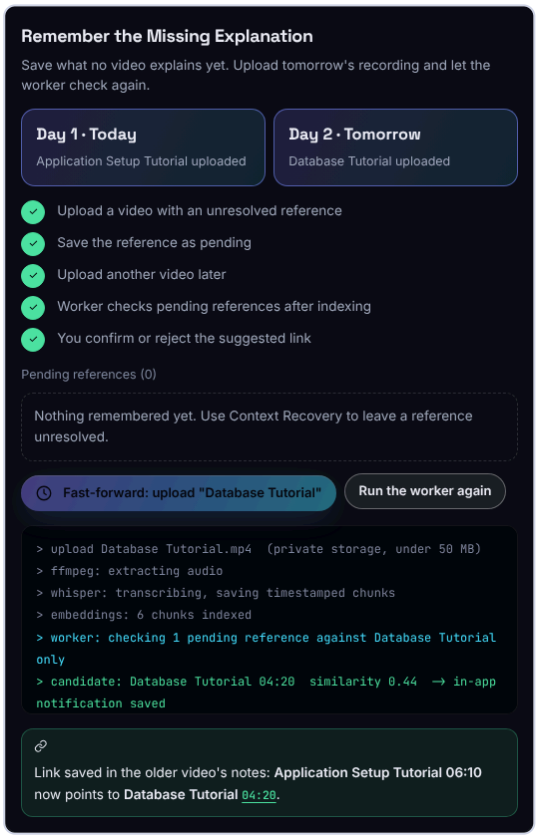


Figure 8.4 · Running the worker again after accepting: nothing pending, no duplicate alert.

CHAPTER 9

Library Map

See which recordings lean on which, what order to watch them in, and what your library assumes but never explains.

9.1 Why it exists

The other three features help one person with one video. The Library Map helps whoever owns the whole library, such as a teacher or a team lead. Every confirmed reference becomes a link between two recordings. Every unresolved reference becomes a gap. Together they answer: *what do my recordings assume viewers already know, and where is it explained?*

9.2 Reading the map



Figure 9.1 · Library Map with one linked reference and one gap.

ELEMENT	MEANING
Solid green line	A confirmed reference. The label shows from-time to source-time, e.g. 09:30 to 03:20.

ELEMENT	MEANING
Dashed amber line	A gap: a reference with no confirmed source. The dashed box reads "Missing explanation, not uploaded yet".
Dashed cyan line	A candidate found by the worker but not yet confirmed.
Linked / Gaps / Explained	Counts of confirmed links and gaps, and the share of references that are explained.

9.3 Suggested watch order

Videos are ordered so that sources come before the videos that depend on them. A video that still has a gap is marked **assumes missing knowledge**. After the Database Tutorial is uploaded and linked, the order becomes Server Setup, Database Tutorial, Application Setup Tutorial.

9.4 Gaps report

Each gap lists the video and timestamp, the sentence, and the topic words to cover, for example `database` `connection` `migrations`. The wording changes to **Possible fix** once a candidate exists. Press **Copy report** to copy a plain-text list you can paste into a message. Some browsers block clipboard access, in which case select the text instead.

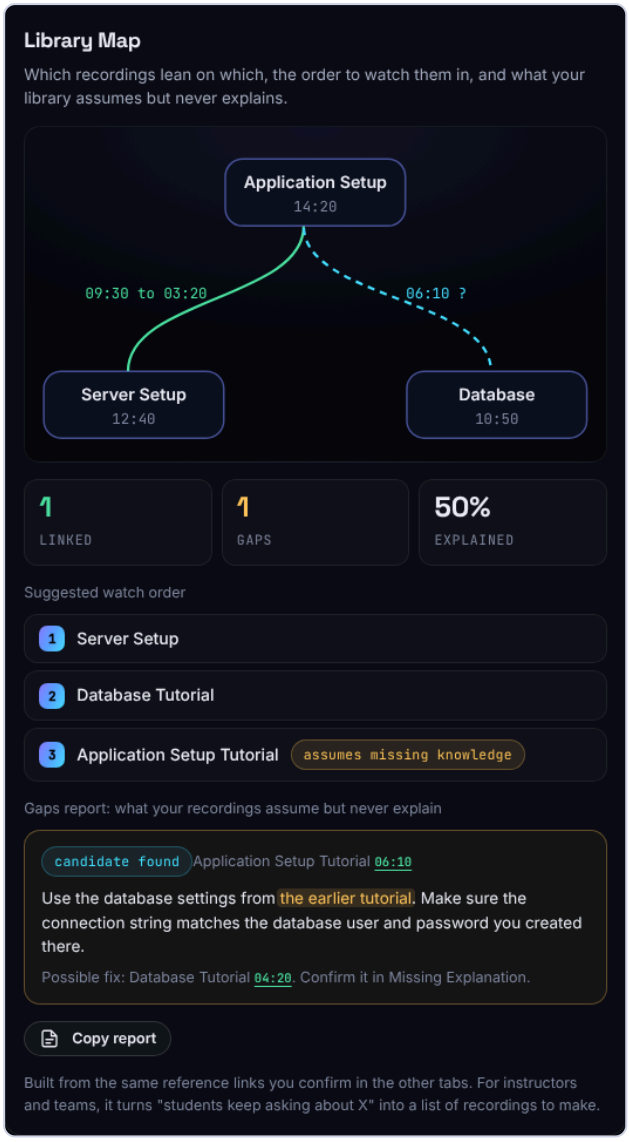


Figure 9.2 · A candidate has appeared: the dashed line turns cyan.

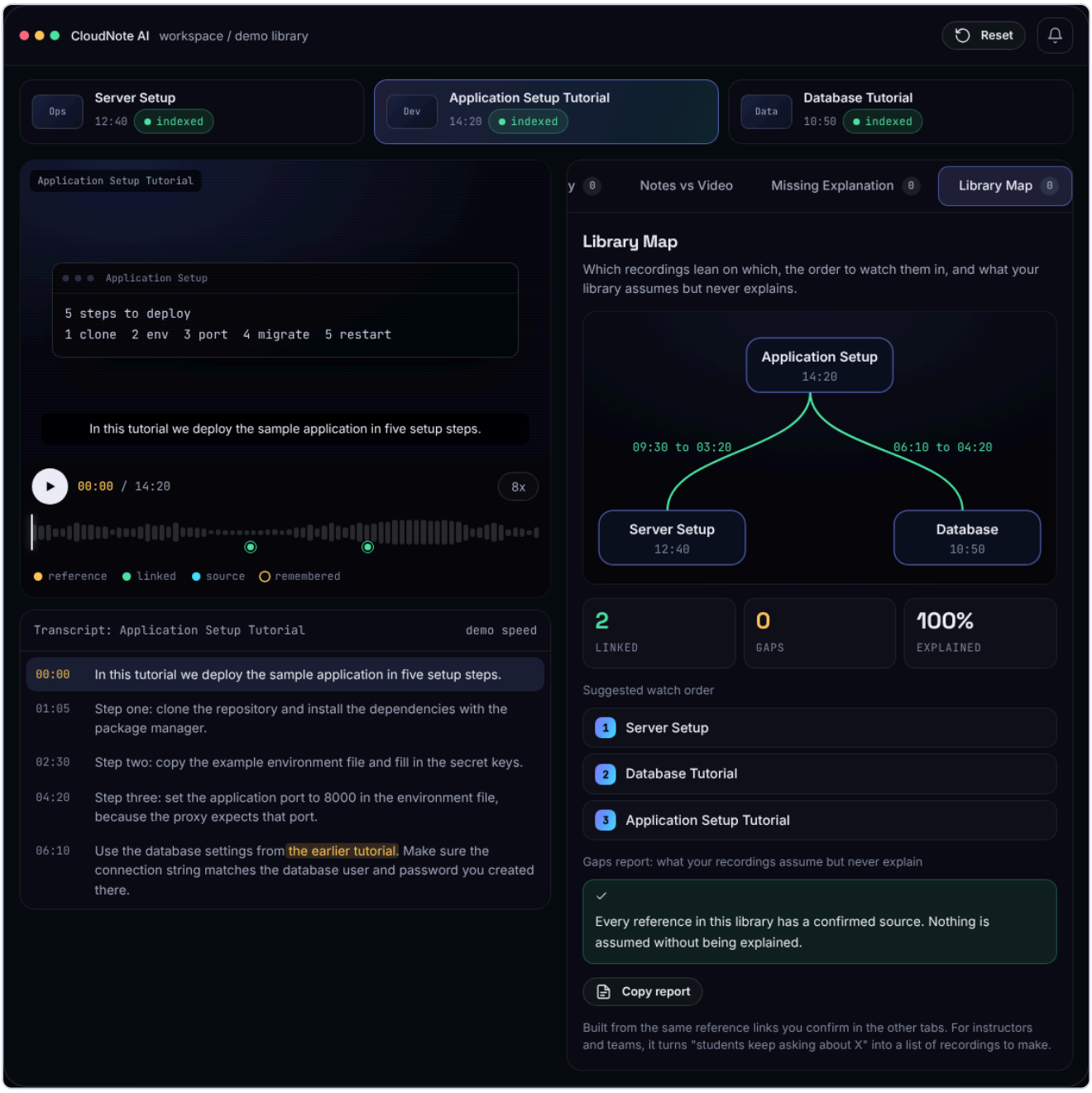


Figure 9.3 · Everything explained: two links, no gaps, 100%.

CHAPTER 10

The 60-second story

A guided run of the whole demo, and a script for presenting it.

10.1 Playing the story

Above the workspace, press **Play the 60-second story**. The demo resets and performs six steps by itself, each with a caption in a blue bar above the library. Any click inside the workspace stops the story and hands control to you.

STEP	CAPTION SUMMARY	WHAT THE DEMO DOES
1	At 06:10 the speaker says "the earlier tutorial"	Opens Context Recovery and seeks the player to 06:10
2	A likely source exists in the library	Accepts the 09:30 reference, linking it to Server Setup 03:20
3	No source yet, so it is remembered	Saves the 06:10 reference as pending
4	Next day, a new video is uploaded	Opens Missing Explanation and uploads the Database Tutorial
5	A possible explanation appears, you decide	Accepts the link
6	The map shows everything explained	Opens the Library Map

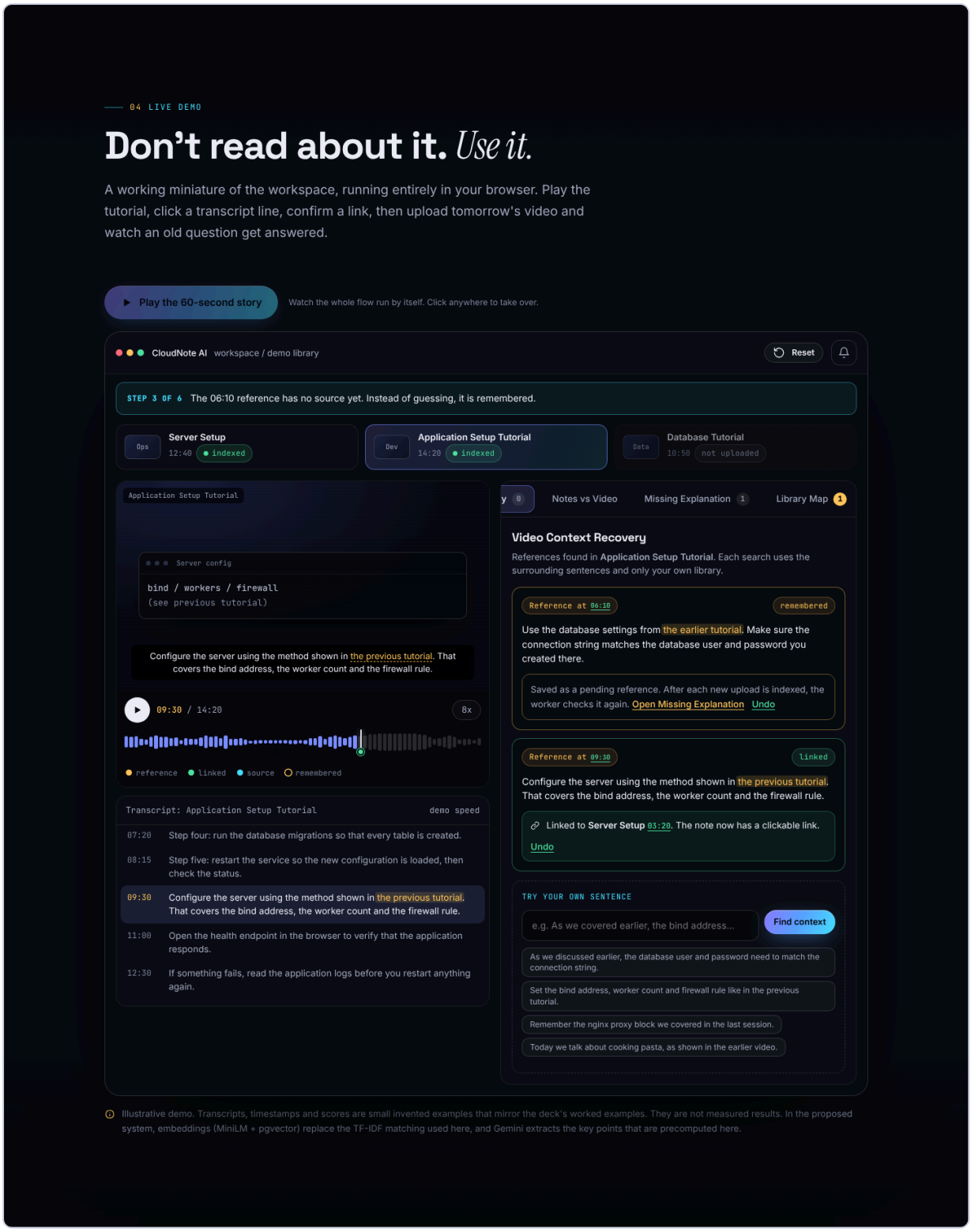


Figure 10.1 · The story running: the caption bar sits above the library strip.

10.2 Three-minute presenter script

1. **Problem (30 s).** "Videos say 'as we covered earlier' and never tell you where. Your notes can drift from the video. Both cost time."
2. **The story (60 s).** Press the button and talk over it: reference detected, one linked, one remembered, tomorrow's upload answers it, the map shows the library is fully explained.

3. **Notes vs Video (45 s).** Compare, point at 8080 versus 8000, fix it in one click. "A wrong port number is exactly what a quick re-read misses."
4. **Library Map (30 s).** "For a teacher or team lead, this is a list of recordings still to make."
5. **Honest close (15 s).** "This is a proposed implementation with illustrative data. Measured evaluation comes next."

Tip: before presenting, press Reset, and open the page on the device you will use to confirm fonts and sound-free playback work as expected.

CHAPTER 11

How it works and Architecture

Two interactive explainers for the technical audience.

11.1 The five-step pipeline

The **How it works** section lists the steps every extension reuses. Click a step, or let it auto-play (about five seconds each, paused once you click). The right pane shows what the step does and an illustrative example of its data.

#	STEP	WHAT HAPPENS
1	Login	Supabase Auth provides the session. FastAPI validates access before accepting a processing request.
2	Store	An MP4 under 50 MB goes to private Supabase Storage. PostgreSQL keeps owner, path and status.
3	Transcribe	A local worker uses FFmpeg to extract audio and Whisper to transcribe it. Text is stored with start and end times. Failed jobs keep an error status and can be retried.
4	Notes	Gemini receives chunks with source IDs and returns key points and a summary. Returned IDs are validated, and timestamps come from the database, never from the model.
5	Search	Local Sentence Transformers (all-MiniLM-L6-v2) encode chunks and questions. Supabase pgvector finds similar chunks among those the user is permitted to see.

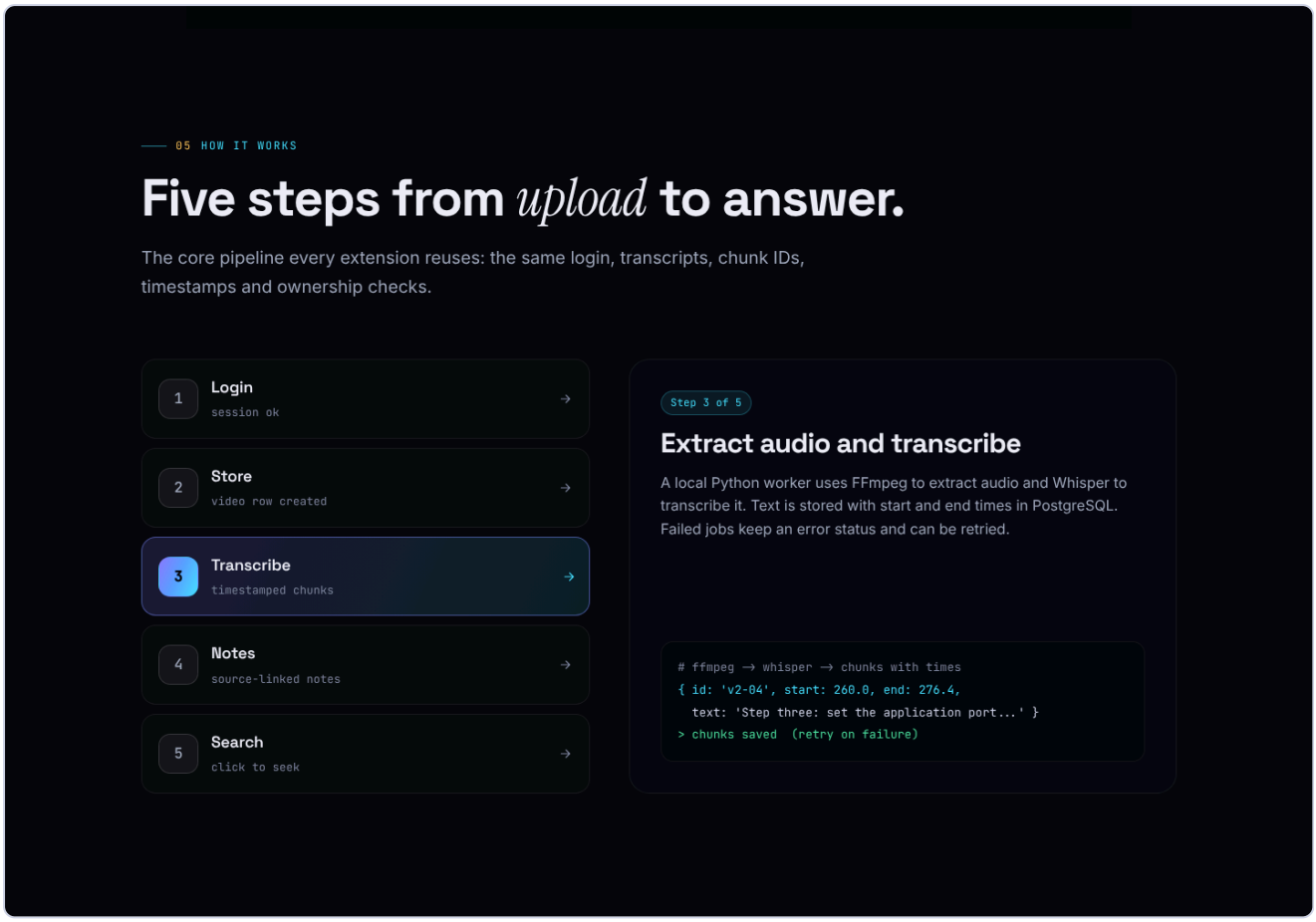


Figure 11.1 · Step 3, Transcribe, selected.

11.2 The architecture explorer

Ten components sit in three zones: **Your laptop** (React, FastAPI, FFmpeg, Whisper, Sentence Transformers), **Supabase** (Auth, Private Storage, PostgreSQL, pgvector) and **Google** (Gemini). Choose one of four flows with the chips above the diagram: *Ingest a video*, *Search or Context Recovery*, *My Notes vs Video* and *Missing Explanation*.

- Components in the chosen flow light up in amber, numbered in order. The cyan outline marks the current step, which advances automatically.
- Click a numbered step in the list to jump to it. Hover or click any component to read what it does. Click again to release it.
- Open **Proposed schema sketch** for a table design (videos, transcript chunks, pending references, notifications). It is a sketch, not final.

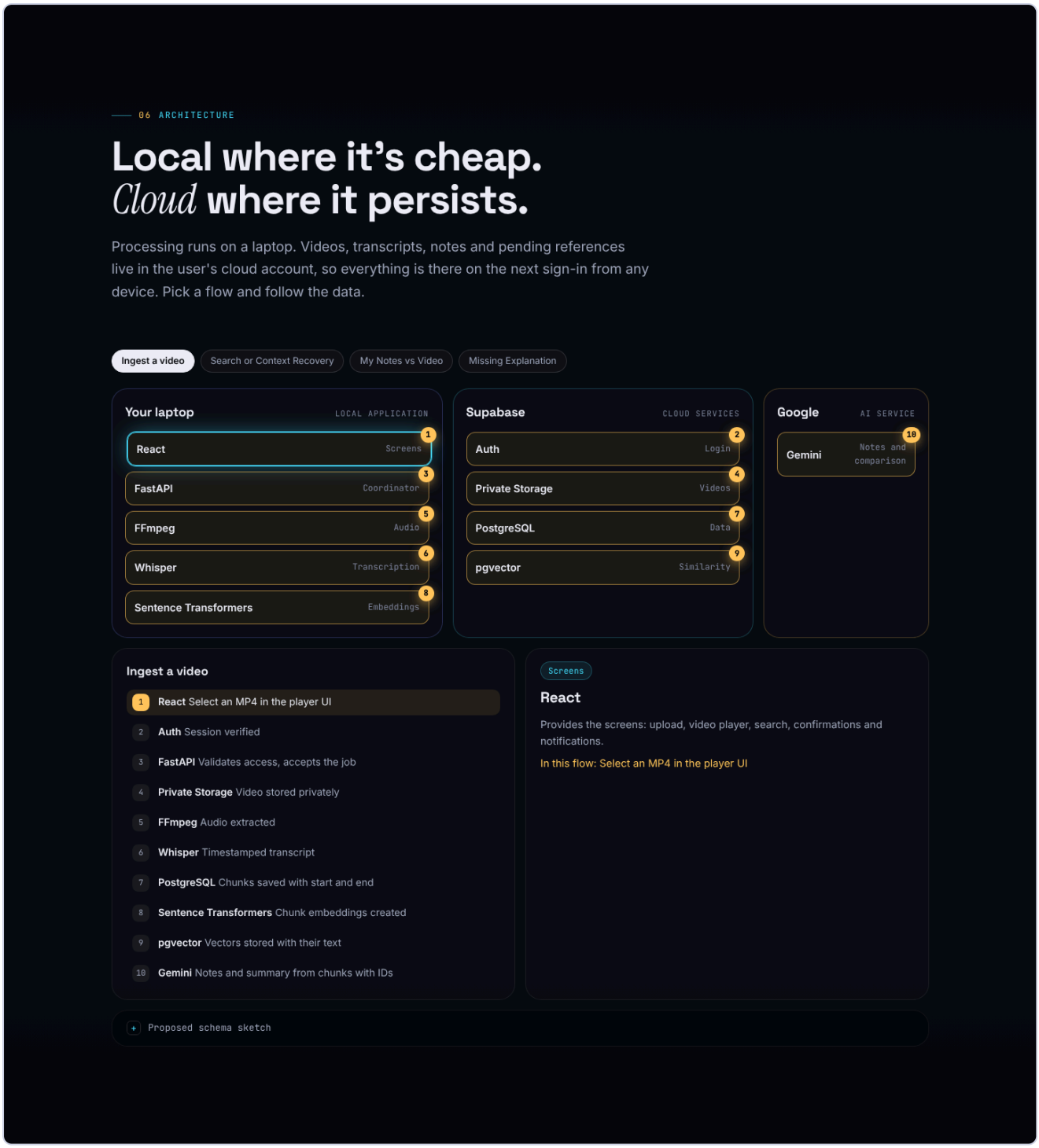


Figure 11.2 · Architecture explorer showing the Ingest a video flow.

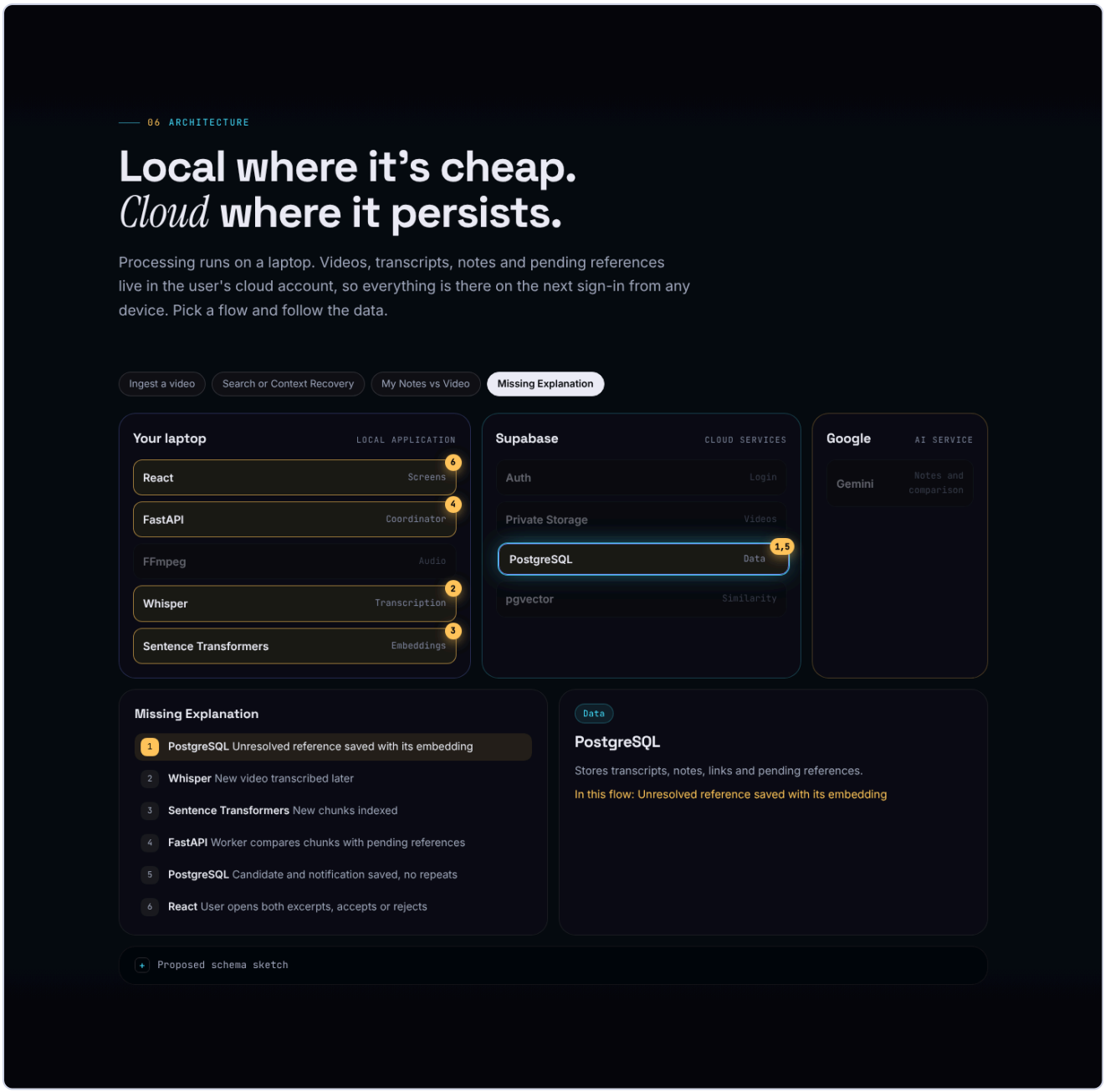


Figure 11.3 · The Missing Explanation flow, with PostgreSQL selected.

- Proposed schema sketch

```
-- Proposed schema sketch (illustrative, not final)
create table videos (
  id uuid primary key,
  owner uuid references auth.users,
  title text, storage_path text,
  status text  -- queued | processing | ready | failed
);

create table transcript_chunks (
  id text primary key,          -- source id, validated
  video_id uuid references videos,
  start_s real, end_s real,
  text text,
  embedding vector(384)         -- all-MiniLM-L6-v2
);

create table pending_references (
  id uuid primary key,
  video_id uuid, chunk_id text, -- where it was said
  surrounding text,
  query_embedding vector(384),
  status text                   -- pending | resolved | rejected
);

create table notifications (
  id uuid primary key,
  reference_id uuid references pending_references,
  candidate_chunk text,
  unique (reference_id, candidate_chunk)  -- no repeat alerts
);
```

Figure 11.4 · The proposed schema sketch.

CHAPTER 12

Research

The literature survey and the research gap.

12.1 The papers

Fifteen papers from 2020 to 2024 cover summarization (BART, PEGASUS, Longformer, SummN, QMSum), retrieval (DPR, ColBERT, ColBERTv2, BEIR, Fusion-in-Decoder, RAG), embeddings (SimCSE), speech (Whisper, WhisperX) and grounded generation (Self-RAG). Each card shows year, venue, technique, data, what was measured, the takeaway, a DOI link and an Open paper button.

- Use the theme chips (Speech, Summarization, Retrieval, Embeddings, Grounding, Benchmarks) to filter. A paper can belong to more than one theme.
- Type in the search box to filter by any word in the title, authors, technique, data or takeaway. All words must match.
- The count under the list shows how many papers are shown.

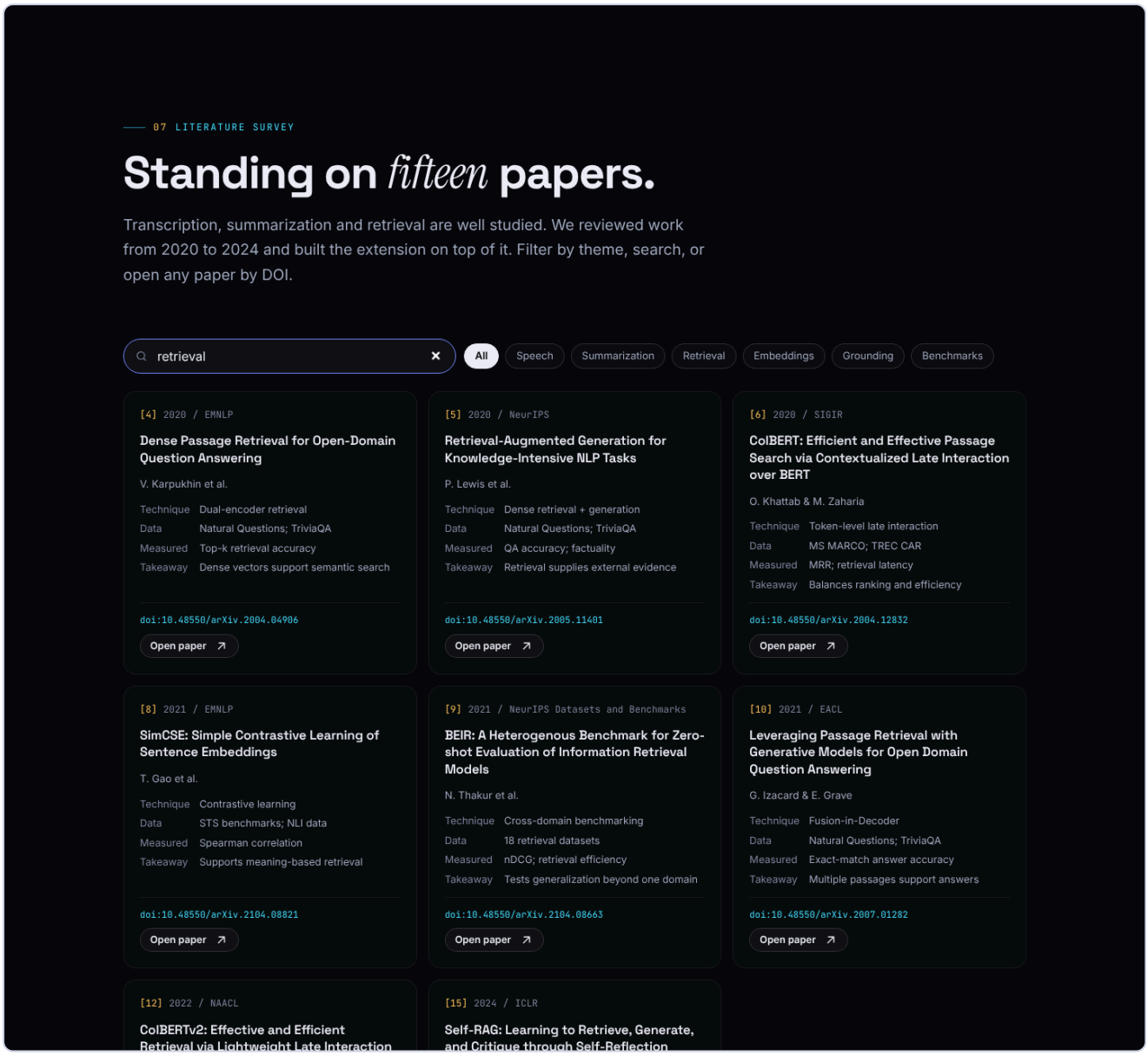


Figure 12.1 · Filtering the survey with the word "retrieval".

12.2 The research gap

A five-row table shows what the project borrows and what it extends.

PAPER	METHOD	KEY FINDING	PROJECT EXTENSION
WhisperX [14]	ASR + alignment	Time-linked text	Link to a second video
SummN [11]	Staged summaries	Long-input handling	Flag unresolved references
DPR [4]	Dense retrieval	Relevant passage search	Retrieve context candidates
RAG [5]	Retrieve + generate	Evidence for answers	Show supporting timestamps
SimCSE [8]	Sentence embeddings	Meaning-based matching	Rank candidate explanations

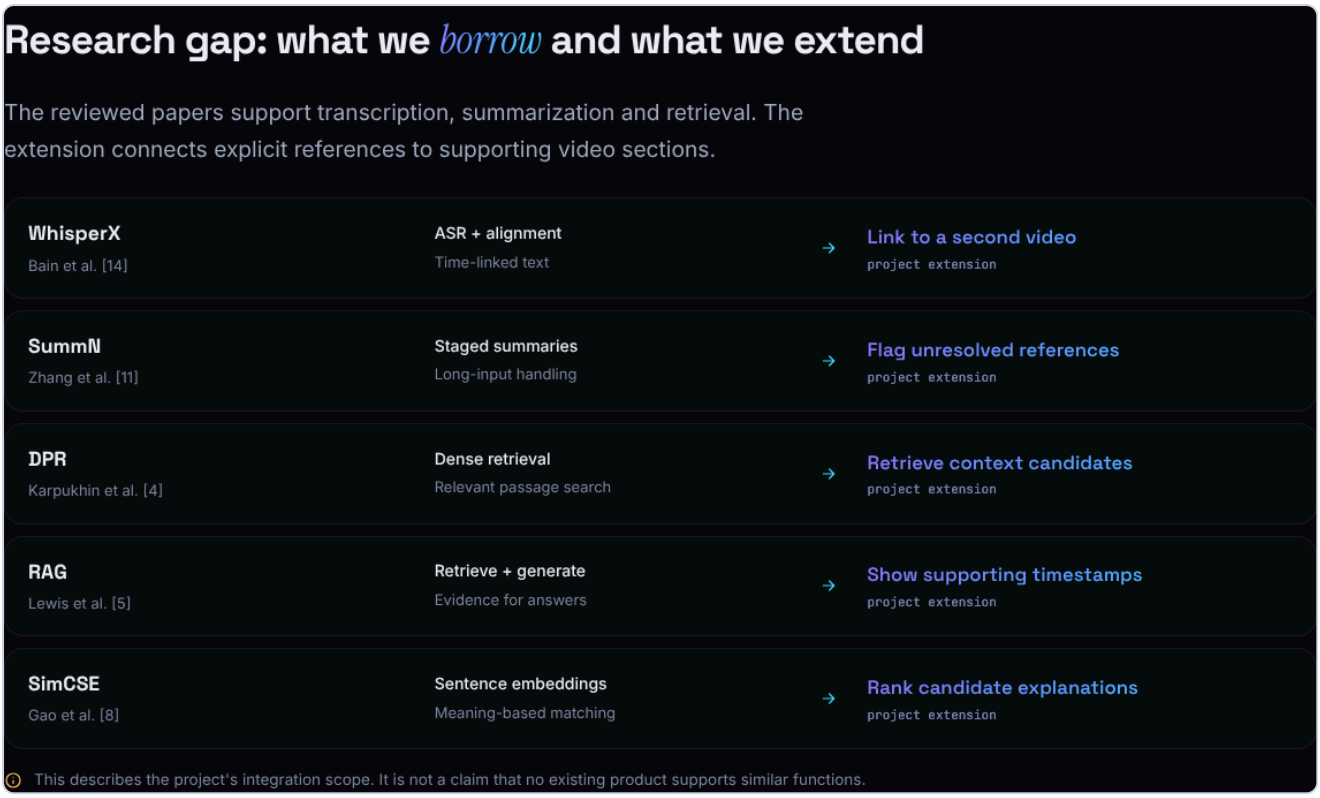


Figure 12.2 · The research gap rows on the site.

> **How to describe the gap**
>
> This is a project-specific integration of existing techniques. It is not a claim that no existing product supports similar functions, and global novelty is not established.

CHAPTER 13

Plan, evaluation and costs

What will be built, how it will be measured, and what the free-tier demo can and cannot do.

13.1 Implementation plan

The estimates apply only once the core pipeline works: login, uploads, transcription, stored chunks and search.

WORK ITEM	ESTIMATE	INCLUDES
Video Context Recovery	3 to 4 days	Phrase detection, candidate search, confirmation controls, basic evaluation
My Notes vs Video	2 to 3 more days	Pasted notes, Gemini comparison, source validation, approved edits
Remember the Missing Explanation	1 to 2 more days	Pending records, checks after indexing, de-duplicated in-app alerts
Total	About 6 to 9 days	For the three extensions, subject to team experience

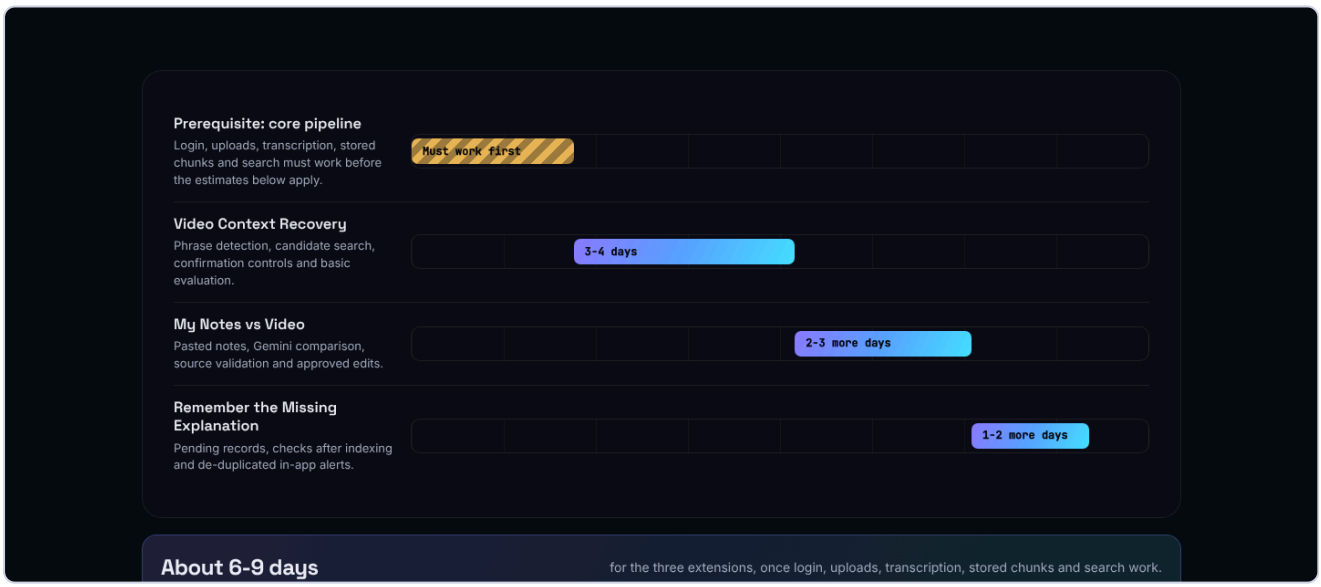


Figure 13.1 · The plan section: bars animate in when scrolled into view.

13.2 Evaluation

AREA	WHAT WILL BE CHECKED
Cases	Clear and ambiguous references, no match, later matching uploads, complete notes, omissions and conflicting details
Checks	Source timestamps, useful top-three suggestions, comparison labels, duplicate alerts, access isolation and review time
Demo scope	3 to 5 short English videos and pasted personal notes, searching only the signed-in user's collection
Limits	Transcript errors affect every feature. Visual-only explanations and missing-audio reconstruction are out of scope

13.3 Free-tier costs and constraints

SERVICE	LIMITS TO KNOW
Supabase Free	500 MB database, 1 GB file storage, 50 MB maximum file size. Free projects may pause after a week of inactivity.
Gemini free tier	Within account limits. Free-tier content may be used to improve Google products, so use non-sensitive demo recordings.
Local processing	React, FastAPI, FFmpeg, Whisper and embeddings run on the laptop. Processing stops while the laptop is off.
Cost boundary	Target of zero service fees within free quotas. Hardware, internet and electricity are excluded. A larger always-online deployment may cost money.

These limits were checked on 18 September 2026. Verify them again before deployment.

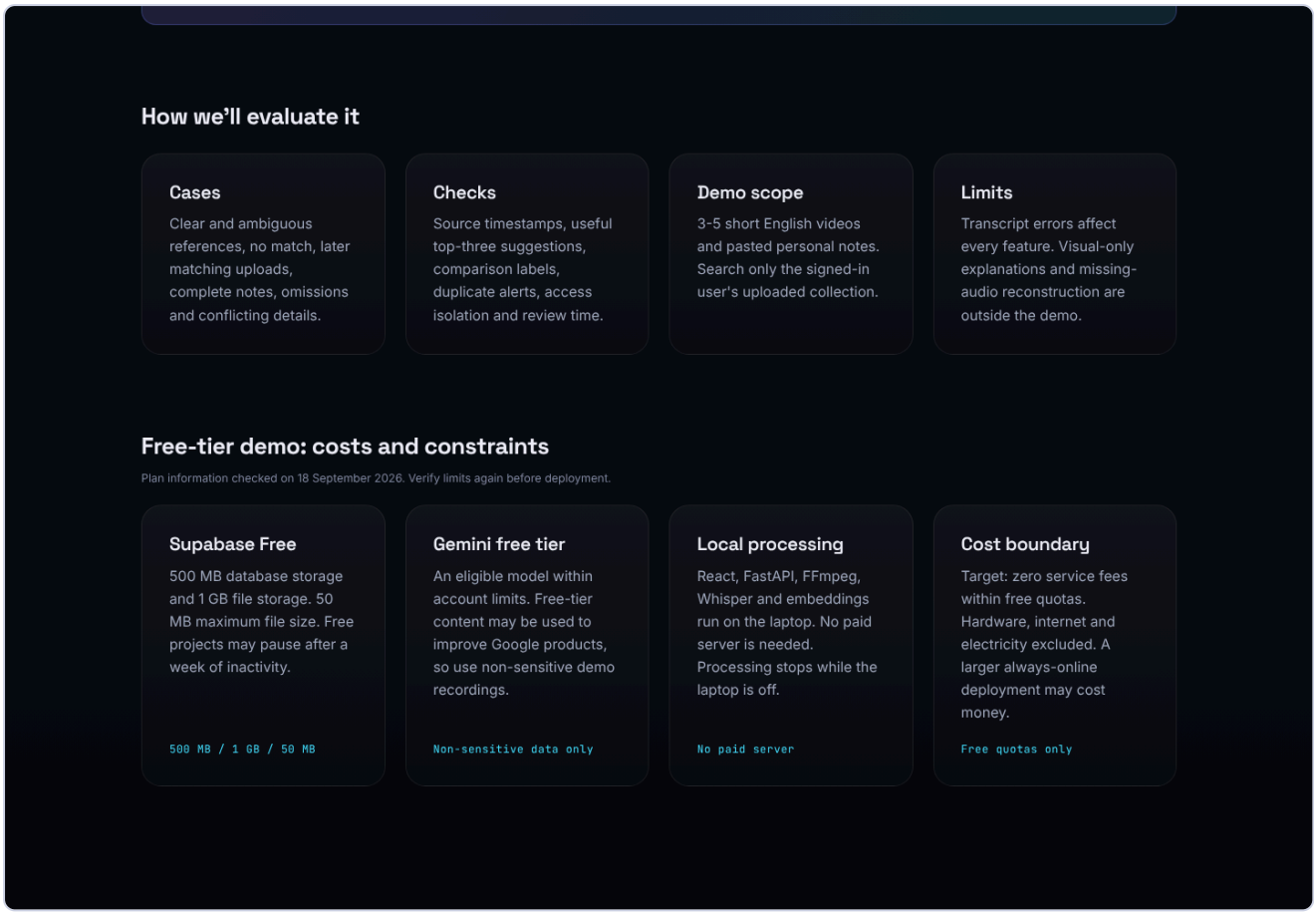


Figure 13.2 · Evaluation and cost cards on the website.

CHAPTER 14

Data and methodology of the demo

Exactly how the in-browser demo finds references, ranks excerpts and compares notes, and how that differs from the proposed system.

14.1 Demo versus proposed system

STEP	IN THE WEBSITE DEMO	IN THE PROPOSED SYSTEM
Transcript	Three hand-written transcripts, 26 chunks	Whisper transcribes uploaded video, with FFmpeg for audio
Embedding	TF-IDF term vectors	all-MiniLM-L6-v2 sentence embeddings (384 dimensions)
Similarity search	Cosine similarity in the browser	pgvector search in Supabase
Reference detection	Phrase rules (below)	Python phrase rules
Key points for notes	Precomputed: five points with chunk IDs	Gemini extracts them, FastAPI validates IDs
Notes storage	Your browser only	Supabase, per user
Pending-reference worker	Runs in the page after the simulated upload	Local Python worker after indexing

14.2 Reference phrases

A chunk is flagged when it matches any of these patterns (case-insensitive):

- "as we discussed / covered / mentioned / saw / explained earlier / before / previously / last time"
- "the previous / earlier / last / prior" followed by video, tutorial, session, lecture, meeting, class, recording, part, episode or module
- "earlier discussion / explanation / section / walkthrough"
- "we already covered / discussed / explained this / that / it earlier / before / previously"
- "last time"
- "in the first / second video / tutorial / session / lecture"

14.3 Ranking excerpts

1. Text is lower-cased, split into words, stop words are dropped, and words are lightly stemmed. A small synonym map joins pairs such as repo and repository, env and environment, app and application.

2. For a search query, words that only describe the act of referring (earlier, previous, tutorial, shown and similar) are also dropped, so the match is about the topic.
3. Each chunk becomes a TF-IDF vector. The score is the cosine similarity between the query and a chunk.
4. Chunks from the video being viewed are excluded. Scores at or above 0.30 are suggested, up to three.

14.4 Comparing notes

Each key point is scored against each line of your notes in the same way. At 0.34 or above the line is a match. If the key point contains a number (two to five digits) that the matched line does not contain, and the line has its own number, the verdict is *possible mismatch*. Otherwise it is *covered*. With no match the key point is *missed*.

14.5 Checking for pending references

After a video is indexed, each pending reference is searched against that video's chunks only. A candidate at or above 0.30 creates one notification. The pair of reference and candidate is remembered, so it is never alerted twice.

14.6 How the demo is verified

Thirteen automated tests lock the behavior to the worked examples: exactly two references are detected in the application tutorial and none in the others; the 09:30 reference resolves to Server Setup at 03:20; the 06:10 reference has no source on day 1 and resolves to Database Tutorial at 04:20 on day 2; duplicates are suppressed; the notes sample yields 3 covered, 1 mismatch (8080 versus 8000) and 1 missed; fixing the port or adding the restart line changes the verdicts; and empty notes do not crash.

CHAPTER 15

Privacy, security and responsible use

What the website stores, what it never sends, and how to use the proposed system responsibly.

15.1 What the website stores

- The website has no accounts, no analytics, no cookies and no server calls for the demo. Nothing you type leaves your browser.
- One item is saved in your browser's local storage: the notes you save in the Notes vs Video tab. **Reset** deletes it. Private browsing or blocked storage simply means notes are not remembered.
- Fonts and all scripts are served from the same site. No third-party content is loaded.

15.2 Hosting security

Both deployment options send these response headers: a content security policy that allows only same-site scripts, fonts and styles; no framing; no content-type sniffing; a strict referrer policy; and HTTPS-only transport. The S3 option keeps the bucket private behind CloudFront.

15.3 Responsible use of the proposed system

- **Sensitive recordings.** The free Gemini tier may use content to improve Google products. Use non-sensitive recordings for the demo.
- **Ownership isolation.** Searches look only at the signed-in user's own uploads. Isolation is one of the planned evaluation checks.
- **User confirmation.** Links are confirmed by the user and note changes are saved only after approval.
- **Transcript errors.** Mistakes in speech recognition affect every feature. Treat suggestions as pointers to check, not as facts.
- **Scope.** The system finds supporting context in videos you uploaded. It does not reconstruct missing audio or visuals.

CHAPTER 16

Technical reference

For maintainers: project structure, building, testing and deploying.

16.1 Project structure

PATH	CONTENTS
src/index.html	Page markup and icon sprite
src/css/	Design tokens and base styles, components, workspace styles
src/js/content.js	Team, papers, plan, FAQ, architecture and pipeline content
src/js/demo-data.js	Demo videos, 26 transcript chunks, key points, sample notes
src/js/nlp.js	Reference detection, TF-IDF search, notes comparison, pending worker
src/js/workspace.js	The interactive workspace, tabs, Library Map and the guided story
src/js/*.js	Hero animation, background, sections, site search, page behavior
tests/	Engine tests and browser checks
deploy/	S3 + CloudFront template and script, Lambda handler and script
dist/	The built site that is deployed (about 0.7 MB)

16.2 Commands

```
npm install
node build.mjs                    # builds dist/ from src/
node tests/engine.test.mjs        # 13 tests
cd dist && python3 -m http.server 8765
```

16.3 Deploying to AWS

OPTION	COMMAND	NOTES
S3 + CloudFront (recommended)	./deploy/deploy-s3.sh us-east-1	Private bucket, CDN, HTTPS and security headers. Prints the live URL.
Lambda Function URL	./deploy/deploy-lambda.sh us-east-1	Python 3.12 handler serving the files with the same headers.

Both need the AWS CLI configured with credentials (`aws configure`). To redeploy after a change, run the build and the same script again. Test the printed URL after the first deploy: the scripts were prepared and the Lambda handler tested locally, but they have not yet been run on a live AWS account.

16.4 Editing content

- Team and guide: `src/js/content.js` (TEAM, GUIDE).
- Papers, FAQ, plan, costs: the same file.
- Demo transcripts and sample notes: `src/js/demo-data.js` . If you change a transcript, rerun the tests: they encode the expected behavior.

> **Team list**
>
> The submitted abstract lists Hemanth Kumar (2320030150) while the Project Review-1 deck lists Sargu Sai Chetan (2320030348). The site uses the deck. Change TEAM in content.js if this is not right.

CHAPTER 17

Troubleshooting and FAQ

Fixes for the issues most likely to come up.

SYMPTOM	FIX
Page is unstyled or fonts are missing when opened by double-clicking	Open through a web server (the preview command in 16.2) or the deployed URL.
The demo shows old notes or a stale state	Press Reset in the workspace. It also clears saved notes.
Copy report says copy was blocked	Some browsers and embedded viewers block clipboard access. Select the report text instead.
Animations look still	Your device is set to reduce motion. The site respects this on purpose.
The search shortcut does nothing	Click the page first, then press Ctrl+K. Inside a text box, / types a slash, so use Ctrl+K.
The story button does nothing after a click	A click inside the workspace stops it. Press the button again to restart.
Notifications show no badge	Upload the Database Tutorial in the Missing Explanation tab first, with a pending reference saved.
Free-tier numbers look out of date	They were checked on 18 September 2026. Re-check Supabase and Gemini pricing before deploying.

17.1 Frequently asked questions

Is the live demo running real AI models?
It runs real code: reference detection, TF-IDF retrieval, note comparison and the pending-reference worker. The proposed system uses Whisper, MiniLM embeddings with pgvector and Gemini. The transcripts and key points in the demo are small illustrative examples.

Are the numbers measured results?
No. The project is at Project Review-1. Measured evaluation comes after a small English-video demo is implemented and tested.

Does it reconstruct missing audio or visuals?
No. It finds supporting context in videos you uploaded.

Who decides whether a suggested link is right?
The user. A score marks a candidate, not proof.

Is this globally novel?
We do not claim that. It is a project-specific integration of existing techniques.

What does the demo cost to run?

The target is zero service fees within free quotas, with processing on a laptop.

APPENDIX A

Demo data

The three illustrative videos and the key moments used in the walkthroughs.

VIDEO	TIME	SAID IN THE VIDEO (SHORTENED)	ROLE
Server Setup	03:20	To configure the server, set the bind address, the worker count and the firewall rule.	Source for the 09:30 reference
Application Setup Tutorial	04:20	Set the application port to 8000 in the environment file.	Key point 3 (notes say 8080)
Application Setup Tutorial	06:10	Use the database settings from the earlier tutorial.	Reference with no source on day 1
Application Setup Tutorial	08:15	Restart the service so the new configuration is loaded.	Key point 5 (missing from the notes)
Application Setup Tutorial	09:30	Configure the server using the method shown in the previous tutorial.	Reference resolved to Server Setup 03:20
Database Tutorial	04:20	Create a database user and password, then the application database.	Answers the 06:10 reference on day 2

Sample notes and key points

KEY POINT	SOURCE	SAMPLE NOTES LINE	VERDICT
Clone the repository and install the dependencies	01:05	1. Clone the repo and install dependencies	Covered
Copy the example environment file and add the secret keys	02:30	2. Copy the example env file and add secret keys	Covered
Set the application port to 8000 in the environment file	04:20	3. Set the app port to 8080 in the env file	Possible mismatch
Run the database migrations to create the tables	07:20	4. Run database migrations	Covered
Restart the service to load the new configuration and check the status	08:15	(none)	Missed

All content is invented for illustration and mirrors the worked examples in the project deck.

APPENDIX B

Team and references

Project details as submitted.

ITEM	DETAIL
Title	CloudNote AI: Cloud-Based Intelligent Video Note Extraction and Semantic Search Platform
Course	Engineering Capstone Project - 1 (23IE4053R/23IE4053A), A.Y. 2026 - 2027
Department	Computer Science and Engineering, Koneru Lakshmaiah Education Foundation
Team	K. Mithilesh Reddy (2320030330), Nithin (2320030186), Sargu Sai Chetan (2320030348)
Project guide	Dr. Neelima Gogineni

Implementation references

- Supabase pricing: supabase.com/pricing
- Supabase storage file limits: supabase.com/docs/guides/storage/uploads/file-limits
- Supabase pgvector: supabase.com/docs/guides/database/extensions/pgvector
- Gemini API pricing: ai.google.dev/gemini-api/docs/pricing
- Sentence Transformers models: sbert.net/docs/sentence_transformer/pretrained_models.html
- Whisper: github.com/openai/whisper · FFmpeg: ffmpeg.org/ffmpeg.html · FastAPI: fastapi.tiangolo.com

The 15 papers surveyed, with DOIs, are listed in the References section of the website.

APPENDIX C

Quick reference

The most used controls on one page.

I WANT TO...	DO THIS
Search the whole site	Ctrl/Cmd + K, or / outside a text box
See the whole flow in a minute	Demo section, Play the 60-second story
Link a reference to its source	Context Recovery tab, Accept this source
Keep a reference for later	Leave unresolved and remember it
Check my notes against the video	Notes vs Video tab, Compare with transcript
Fix a wrong number in my notes	Use the video's value, then Save notes
Simulate tomorrow's upload	Missing Explanation tab, Fast-forward
See what is not explained	Library Map tab, Gaps report, Copy report
Start over	Reset (top right of the workspace)
Seek precisely	Focus the waveform: arrows, Shift+arrows, Home, End, Space
Read a paper	Research section, Open paper or the DOI link
Deploy	./deploy/deploy-s3.sh us-east-1

CHIP	MEANING
Covered	Your notes contain the point
Possible mismatch	A number differs from the video
Missed key point	Not in your notes
needs a source	Reference without a confirmed source
linked	Reference with a confirmed source
remembered	Pending, checked after each upload
candidate found	Suggested source awaiting your decision

GLOSSARY

Glossary

Terms used in this manual.

TERM	DEFINITION
ASR	Automatic speech recognition: converting speech to text.
Chunk	A short transcript segment with a start time and an ID.
Cosine similarity	A score from 0 to 1 for how closely two vectors point in the same direction.
CSP	Content security policy: a header limiting what a page may load.
DOI	Digital object identifier: a permanent link to a paper.
Embedding	A list of numbers representing the meaning of a piece of text.
FFmpeg	A tool for processing audio and video, used to extract audio.
Gap	A reference with no confirmed source in the library.
Gemini	Google's AI model, used to write notes and compare notes with the transcript.
IDF	Inverse document frequency: gives rarer words more weight.
MiniLM	A small sentence-embedding model (all-MiniLM-L6-v2) used in the proposed system.
Pending reference	A saved reference waiting for a future upload to explain it.
pgvector	A PostgreSQL extension for storing and searching embeddings.
RAG	Retrieval-augmented generation: answering from retrieved evidence.
Supabase	The cloud service for login, file storage and the database.
TF-IDF	A way to weigh words by how often they appear in a text and how rare they are across texts.
Threshold	The minimum score for a suggestion. 0.30 in the demo.
Whisper	OpenAI's speech-recognition model, run locally in the proposed system.